

Summary of public comments

Considerations for the Regulation of Generative AI-Enabled Medical Devices
Docket FDA-2026-N-7874

124 tagged comments, 2398 tags, 2139 skips, 5 duplicates not tagged · 40 sections · generated 02 October 2026

Each section says what commenters wrote about one part of FDA's paper. The line under a section's heading counts the comments and tags behind it, and the positions those tags take.

A number in brackets, such as [3], points to that section's sources, listed under its text. A source names a comment and a paragraph: 0126 ¶p091 is paragraph p091 of comment FDA-2026-N-7874-0126, so every claim can be looked up in the commenter's own words.

GENERAL: Overall views, background and definitions

84 comments, 284 tags: 48 support, 202 conditional, 8 oppose, 26 neutral

GEN: Overall approach and general views

79 comments, 224 tags: 41 support, 171 conditional, 7 oppose, 5 neutral

Supporters welcome FDA seeking input before setting policy [1], and many endorse its core pieces, such as the competency approach, the two-axis risk framework or a risk-based lifecycle approach, often calling them sound or a strong foundation [2]. Some agree that static, point-in-time validation cannot keep up with GenAI's variable, changing behavior, at least for some devices [3]. Several want premarket evaluation carried into continuing lifecycle oversight [4], with authorization treated as an initial rather than a permanent finding [5].

Several ask FDA to build on machinery already in use, such as ISO 14971 risk files, design, supplier and change controls, and drug-side tools, rather than a parallel system [6]. Others seek clarity and lower burden: a pathway decision matrix, CMS coordination and a small-business pathway with fee relief [7], streamlined pathways for clinician-supervised tools [8], and fixed compliance costs that don't entrench incumbents [9]. Several urge international harmonization with partner regulators, including through IMDRF [10]. Others want FDA to say which proposals need new legal authority or which existing mechanisms apply [11], to settle how a licensure-style scheme fits state regulation of medical practice [12], and to coordinate with other agencies, including on AI that influences care outside the device definition [13].

Some caution that flexibility should not outrun the infrastructure that makes it safe, such as trustworthy benchmarks and monitoring proven reliable before it earns reduced premarket evidence [14]; that least-burdensome goals must not displace reasonable assurance of safety and effectiveness [15]; that pilot or sandbox flexibility should rest on robust premarket evidence [16]; and that the framework must be able to conclude an intended use is not yet supportable [17]. One commenter finds the paper's diagnosis stronger than its solutions and warns against "competency" becoming a kind of surrogate biomarker [18].

Several insist human oversight be real, not nominal [19]: one wants an accountable, licensed human to sign consequential decisions while routine automation stays light [20], and others want claimed review spelled out, such as who reviews what, with what expertise and authority [21]. Others want accountability that tracks control and can be proven [22]. Several want the deployed system, not the model alone, evaluated [23]. Individual commenters propose deterministic checks around the model [24], versioned assurance cases or runtime evidence packages [25], revalidation before acting after material change [26], a reproducibility floor beside every performance estimate [27], an independence check on layered safeguards for higher-risk devices [28], and a "Clinical AI License" with autonomy earned in stages [29].

Others name gaps: operation, servicing, cybersecurity and outage safety, with hospital engineering staff involved [30]; settings with no reachable clinician [31]; a pediatric pathway [32]; systemic and subjective bias [33]; and non-generative AI, with autonomy as a cross-cutting topic [34]. A few want patients, clinicians or the public involved earlier, in designing products or shaping policy [35].

The seven opposing points come from a few commenters: a cancer patient and a psychiatry resident who prefer human care and see GenAI as too risky [36], one who would bar use until open questions are settled [37], one who rejects experimental use without consent and opts out [38], and one who argues the

reverse, that AI should get room and be left to market forces [39].

Sources. [1] 0003 ¶p009; 0013 ¶p185; 0024 ¶p004; 0035 ¶p006; 0100 ¶p005; 0126 ¶p297 [2] 0003 ¶p088; 0006 ¶p018; 0012 ¶p001; 0030 ¶p001; 0035 ¶p129; 0045 ¶p126; 0048 ¶p001; 0049 ¶p118; 0051 ¶p009; 0087 ¶p048; 0093 ¶p009; 0098 ¶p046; 0099 ¶p026; 0100 ¶p077; 0106 ¶p019; 0114 ¶p004; 0124 ¶p020 [3] 0029 ¶p004; 0087 ¶p028; 0102 ¶p001; 0117 ¶p001 [4] 0027 ¶p080; 0035 ¶p007; 0039 ¶p006; 0114 ¶p004; 0115 ¶p004 [5] 0029 ¶p021; 0045 ¶p013 [6] 0013 ¶p176; 0035 ¶p123; 0067 ¶p035; 0087 ¶p050; 0093 ¶p009 [7] 0003 ¶p084, ¶p089 [8] 0012 ¶p012 [9] 0094 ¶p021, ¶p022 [10] 0003 ¶p086; 0024 ¶p010; 0038 ¶p005; 0051 ¶p009 [11] 0013 ¶p182; 0116 ¶p023 [12] 0045 ¶p113 [13] 0025 ¶p407, ¶p481; 0035 ¶p014 [14] 0013 ¶p186; 0008 ¶p016 [15] 0107 ¶p005 [16] 0113 ¶p004 [17] 0013 ¶p184 [18] 0020 ¶p065, ¶p071 [19] 0046 ¶p021; 0054 ¶p020; 0092 ¶p021; 0098 ¶p026 [20] 0046 ¶p007 [21] 0092 ¶p021; 0098 ¶p026 [22] 0017 ¶p016; 0035 ¶p011 [23] 0017 ¶p056; 0032 ¶p111; 0035 ¶p005; 0054 ¶p042; 0087 ¶p050 [24] 0054 ¶p103; 0103 ¶p008 [25] 0035 ¶p008; 0115 ¶p048 [26] 0105 ¶p006, ¶p007 [27] 0095 ¶p003 [28] 0104 ¶p026 [29] 0045 ¶p100, ¶p117 [30] 0032 ¶p001, ¶p002, ¶p022 [31] 0082 ¶p035 [32] 0021 ¶p023 [33] 0004 ¶p003 [34] 0126 ¶p241, ¶p243 [35] 0017 ¶p054; 0038 ¶p006; 0019 ¶p042 [36] 0007 ¶p001; 0019 ¶p003, ¶p012, ¶p016 [37] 0011 ¶p004 [38] 0041 ¶p001 [39] 0031 ¶p017

BKG: GenAI characteristics, risks and benefits

17 comments, 27 tags: 5 support, 2 conditional, 1 oppose, 19 neutral

Several agree the paper rightly names GenAI's challenges, such as open-ended inputs, variable outputs and opaque third-party models [1]; one calls its tone, if anything, too confident for the evidence [2]. Others cite hazards: a reported hospitalization after a chatbot suggested bromide for salt, AI reportedly missing abnormal heartbeats and misidentifying fetal anatomy [3], studies of models repeating or elaborating planted false details [4], and a research agent whose actions changed across identical reruns [5]. One warns unconstrained GenAI puts patients at risk of grievous harm [6]; another calls these devices "hugely problematic" because of biased training data [7]. In surgery, one notes, an output may influence an irreversible act with seconds to catch errors [8]. Hopes include wider access, less burnout, help for young people facing long mental health waits, and clinicians kept anchored to current evidence [9], and better rural stroke outcomes if other hospitals accept AI reads [10].

Sources. [1] 0005 ¶p009; 0020 ¶p011; 0038 ¶p001; 0106 ¶p011 [2] 0106 ¶p011 [3] 0023 ¶p002, ¶p004 [4] 0106 ¶p011 [5] 0095 ¶p022, ¶p028 [6] 0034 ¶p002 [7] 0011 ¶p001 [8] 0017 ¶p002, ¶p018 [9] 0012 ¶p002; 0021 ¶p006; 0025 ¶p076; 0086 ¶p002 [10] 0036 ¶p001

DEF: Definitions, scope and device status

19 comments, 33 tags: 2 support, 29 conditional, 2 neutral

Several want regulatory treatment set by what a function does and who relies on it, not by model architecture, fine-tuning, location or deployer [1]. Some want low-risk informational, non-directive functions treated as non-devices or left under enforcement discretion, consistent with decision-support and wellness policies [2], while one wants the decision-support exemption's four-part test re-examined as arbitrary for modern chatbots [3]. One ties lower risk to a clinician independently reviewing an output's basis [4]; another notes generated outputs generally cannot fall within that exclusion [5]. Others ask where conveying a clinician's plan ends and independent judgment begins [6], or whether insight-generating tools count as generative [7]. Commenters raise particular products: radiologist-supervised pre-drafts under conditioned enforcement discretion [8], AI therapists claiming to treat psychiatric conditions [9], combination products [10], multiple-function platforms [11], point-of-care implants [12] and administrative-seeming patient-facing tools [13]. Some find the paper's scope too narrow, because it sets the rest of the device aside [14] or because upstream risks may escape a device-only view [15], and one suggests consensus definitions across agencies [16].

Sources. [1] 0035 ¶p015, ¶p016; 0080 ¶p012; 0087 ¶p009; 0120 ¶p041 [2] 0080 ¶p005, ¶p010; 0094 ¶p023 [3] 0116 ¶p020 [4] 0046 ¶p016 [5] 0126 ¶p087 [6] 0072 ¶p006 [7] 0069 ¶p001 [8] 0120 ¶p027, ¶p058 [9] 0023 ¶p006 [10] 0067 ¶p013 [11] 0018 ¶p007 [12] 0053 ¶p005 [13] 0025 ¶p020 [14] 0032 ¶p044 [15] 0038 ¶p010, ¶p011 [16] 0038 ¶p009

RISK: Assessing a device's risk

73 comments, 428 tags: 36 support, 382 conditional, 9 oppose, 1 neutral

Q1: Two-axis risk framework and additional dimensions

61 comments, 175 tags: 12 support, 159 conditional, 3 oppose, 1 neutral

Most commenters accept the two axes, often as a useful starting point or heuristic, but say they are incomplete or need refining [1]. Some endorse them outright, as intuitive, sound or well conceived [2]. The three opposing points come from one commenter who says the framework misses GenAI risk and is unnecessary given HIPAA [3], and another who wants standard probability-and-severity analysis instead, since activity is not probability of harm [4].

The additions most often requested are reversibility [5], time pressure or time to harm [6], and traceability of outputs to their sources [7]. Views on form differ: reversibility as its own dimension [8], a gating factor [9], a modifier on the consequence axis [10], or a gate on trading premarket evidence for monitoring, since severity can rank irreversible harms below reversible ones [11]. One calls traceability impossible for opaque models [12]; another warns that attaching citations doesn't by itself make output traceable [13]. Some tie these factors to settings: emergency and intensive care, where time pressure amplifies automation bias [14], anaphylaxis triage [15], and lay-rescuer guidance, where tracing steps to fixed resuscitation protocols matters most [16].

Several want detectability of errors added [17], and others want human review credited by its real quality rather than its nominal presence [18]. One wants a named, accountable reviewer to lower assessed risk [19], while others note supervision may be unverifiable from records or satisfiable on paper [20]. Several want the system's authority to act, such as tool access or execution rights, weighed explicitly [21]. Some would credit downstream safeguards only when validated or binding [22]. Several say "incorrect output" is too narrow, since harm also comes from omission, harmful agreement and answering only the literal question [23].

On structure, several prefer keeping two axes and handling extra factors as modifiers, to preserve usability [24], while others propose three layers, four or five axes, or a multiplicative formula [25]. One would reorganize the activity axis around whether a function presents claims from an identified source or generates its own [26]. Some separate an initial screening or classification from later product-specific risk analysis, or inherent from residual risk [27]. Some want alignment with ISO 14971, existing FDA "model influence" terms, IMDRF categories or HHS's high-impact AI classification [28], and some want activity-axis placement shown by architecture, such as deterministic gates, rather than asserted [29].

Individual commenters add other factors: third-party model dependence and change dynamics [30]; connectivity and fleet-wide common-mode failures [31]; data exposure or sensitivity [32]; the care available without the device [33]; duration or scale of exposure [34]; relational features and how far a user's own assertions sway the device [35]; patient vulnerability [36]; pediatric use [37]; and implants as physical outputs [38].

Sources. [1] 0003 ¶p017; 0005 ¶p011; 0009 ¶p008; 0013 ¶p024; 0015 ¶p001; 0017 ¶p020; 0020 ¶p021; 0025 ¶p026; 0028 ¶p002; 0030 ¶p002; 0035 ¶p039; 0037 ¶p002; 0038 ¶p012; 0043 ¶p002; 0046 ¶p002; 0048 ¶p002; 0049 ¶p012; 0051 ¶p010; 0052 ¶p004; 0054 ¶p054; 0087 ¶p011; 0093 ¶p010; 0094 ¶p033; 0099 ¶p007; 0100 ¶p021; 0103 ¶p011; 0112 ¶p007; 0115 ¶p008; 0118 ¶p007; 0120 ¶p129; 0126 ¶p048; 0127 ¶p002 [2] 0020 ¶p012; 0029 ¶p019; 0049 ¶p024; 0076 ¶p024; 0080 ¶p004; 0095 ¶p047; 0102 ¶p005; 0117 ¶p004 [3] 0019 ¶p003, ¶p020 [4] 0055 ¶p013, ¶p014 [5] 0003 ¶p019; 0009 ¶p009; 0017 ¶p021; 0024 ¶p005; 0028 ¶p003; 0030 ¶p003; 0035 ¶p040; 0039 ¶p002; 0045 ¶p020; 0046 ¶p015; 0048 ¶p002; 0049 ¶p027; 0053 ¶p007; 0054 ¶p054; 0086 ¶p003;

0099 ¶p007; 0100 ¶p023; 0102 ¶p010; 0105 ¶p021; 0113 ¶p010; 0115 ¶p010; 0117 ¶p010; 0118 ¶p007; 0120 ¶p130; 0126 ¶p089; 0131 ¶p024 [6] 0003 ¶p020; 0009 ¶p009; 0015 ¶p001; 0017 ¶p021; 0028 ¶p005; 0030 ¶p003; 0035 ¶p041; 0045 ¶p021; 0049 ¶p027; 0054 ¶p054; 0082 ¶p003; 0086 ¶p003; 0105 ¶p021; 0113 ¶p010; 0115 ¶p010; 0118 ¶p010; 0119 ¶p028; 0120 ¶p130; 0126 ¶p089; 0131 ¶p029 [7] 0009 ¶p011; 0015 ¶p003; 0028 ¶p006; 0035 ¶p043; 0049 ¶p027; 0054 ¶p054; 0082 ¶p017; 0087 ¶p014; 0088 ¶p035; 0093 ¶p010; 0094 ¶p033; 0100 ¶p021; 0102 ¶p010; 0105 ¶p021; 0111 ¶p022; 0113 ¶p010; 0115 ¶p011; 0117 ¶p010; 0118 ¶p010; 0120 ¶p130; 0126 ¶p066; 0131 ¶p029 [8] 0003 ¶p019; 0118 ¶p007 [9] 0099 ¶p007 [10] 0028 ¶p021; 0126 ¶p089 [11] 0131 ¶p024, ¶p026 [12] 0043 ¶p002 [13] 0126 ¶p086 [14] 0028 ¶p005 [15] 0118 ¶p010 [16] 0082 ¶p017 [17] 0013 ¶p026; 0020 ¶p022; 0024 ¶p005; 0030 ¶p003; 0039 ¶p002; 0043 ¶p002; 0048 ¶p002; 0051 ¶p010; 0115 ¶p012 [18] 0009 ¶p010; 0017 ¶p021; 0045 ¶p022; 0087 ¶p012; 0098 ¶p028 [19] 0046 ¶p015 [20] 0042 ¶p005; 0119 ¶p027 [21] 0017 ¶p021; 0045 ¶p019; 0049 ¶p027; 0115 ¶p009 [22] 0028 ¶p004; 0052 ¶p024; 0127 ¶p030 [23] 0021 ¶p009; 0025 ¶p045; 0055 ¶p020; 0111 ¶p009 [24] 0013 ¶p026; 0028 ¶p007; 0035 ¶p047; 0049 ¶p024; 0051 ¶p010; 0055 ¶p021; 0087 ¶p015 [25] 0015 ¶p003; 0020 ¶p062; 0038 ¶p014; 0045 ¶p016; 0126 ¶p060 [26] 0126 ¶p066, ¶p067 [27] 0030 ¶p004; 0035 ¶p039; 0052 ¶p009 [28] 0013 ¶p027; 0052 ¶p007; 0067 ¶p017; 0094 ¶p035; 0126 ¶p056, ¶p057 [29] 0103 ¶p012; 0127 ¶p028 [30] 0003 ¶p018; 0093 ¶p010 [31] 0032 ¶p020, ¶p052 [32] 0076 ¶p032; 0101 ¶p004 [33] 0006 ¶p019; 0082 ¶p016; 0094 ¶p034 [34] 0006 ¶p019; 0020 ¶p023 [35] 0021 ¶p022; 0111 ¶p011 [36] 0024 ¶p005; 0039 ¶p002; 0047 ¶p002 [37] 0021 ¶p023 [38] 0053 ¶p007

Q2: Directiveness spectrum for informational functions

31 comments, 50 tags: 11 support, 37 conditional, 2 oppose

Many agree with FDA that directiveness depends on an output's substance and context rather than its wording, and that disclaimers such as "talk to your doctor" don't by themselves reduce it [1]. Several want predictability through worked examples, anchored rubrics, decision trees or tables [2], and one proposes a rebuttable presumption that outputs with specific numbers, such as doses, are action-directing [3]. Modifiers that several name include specificity, personalization, urgency and omitted alternatives [4]. Some would judge directiveness by its effect, such as what a user is likely to do, how often clinicians override outputs, or pressure built up over an exchange [5], and one notes that cost, coverage or provider-ranking information can steer care [6].

Others want directiveness fixed by design: a declared ceiling with a measured escape rate [7], assessment by defined output classes [8], or a scale from presenting options to executing actions, with a rung for omitting options [9]. Of the two opposing points, one rejects grading by wording in favor of a general versus patient-specific split [10], and the other rejects treating directiveness as a property of the function, since in its record it varied with user displeasure [11]. For emergencies, some note that directive output is often the safer design [12].

Sources. [1] 0006 ¶p020; 0020 ¶p013; 0028 ¶p022; 0038 ¶p017; 0049 ¶p030; 0055 ¶p030; 0076 ¶p024; 0082 ¶p018; 0093 ¶p011; 0100 ¶p024; 0113 ¶p012; 0118 ¶p012; 0120 ¶p137; 0126 ¶p049, ¶p051; 0127 ¶p020 [2] 0006 ¶p020; 0028 ¶p022; 0051 ¶p011; 0093 ¶p011; 0100 ¶p025 [3] 0003 ¶p022 [4] 0006 ¶p020; 0045 ¶p025; 0048 ¶p003; 0049 ¶p030; 0052 ¶p033; 0093 ¶p011; 0120 ¶p136 [5] 0005 ¶p251; 0049 ¶p031; 0058 ¶p037; 0113 ¶p012 [6] 0025 ¶p059 [7] 0013 ¶p031 [8] 0100 ¶p026 [9] 0131 ¶p033, ¶p040 [10] 0055 ¶p027, ¶p029 [11] 0065 ¶p031 [12] 0082 ¶p018; 0100 ¶p027

IV-ACT: Action-taking functions (no numbered question)

16 comments, 27 tags: 27 conditional

Most commenters say human supervision should count as lowering risk only when it is real, such as a reviewer with the competence, information, time and authority to override [1]; some want sponsors to prove it works, including against automation bias [2]. Some want review enforced by design: high-risk outputs held for expert review, sign-off for consequential steps, or review that can't be bypassed [3], and one wants output a clinician approves before patients see it placed lower on the activity axis [4]. Others want AI never used without human oversight, or diagnosis, prescribing, treatment planning and triage always approved by a licensed clinician [5], while one wants supervised tools such as scribes and second reads kept in lower-risk tiers [6]. One proposes a five-level authority scale [7]; another ties safeguards to a function's risk profile, not to whether it is labeled autonomous [8].

Sources. [1] 0005 ¶p259; 0017 ¶p047; 0021 ¶p016; 0025 ¶p400; 0055 ¶p031; 0058 ¶p034; 0092 ¶p120; 0099 ¶p013; 0120 ¶p050; 0126 ¶p220 [2] 0021 ¶p016; 0055 ¶p031; 0099 ¶p013 [3] 0092 ¶p006, ¶p043; 0120 ¶p133 [4] 0129 ¶p007 [5] 0023 ¶p006; 0047 ¶p003 [6] 0012 ¶p008 [7] 0017 ¶p022 [8] 0126 ¶p221

IV-MEAS: Measurement, signal-processing and IVD functions (no numbered question)

3 comments, 4 tags: 2 support, 2 conditional

The two supporting points agree with FDA that measurement, signal-processing and IVD functions may carry higher risk because users can't check their outputs or the basis for them [1]. One commenter contrasts such outputs with a private text pre-draft that a radiologist checks against the full source images, calling that verification a material risk control [2]. Another wants detection addressed explicitly as a function type, since a missed imaging finding gives no error signal [3].

Sources. [1] 0049 ¶p025; 0126 ¶p053 [2] 0120 ¶p043, ¶p044, ¶p047 [3] 0126 ¶p169, ¶p171

Q3: Patient-facing vs clinician-facing functions

31 comments, 44 tags: 5 support, 37 conditional, 2 oppose

Many commenters say patient-facing status should not by itself raise a function's risk, often pointing instead to what an output could lead a patient to do or which safeguards are in place [1]. Some note that the realistic alternative is often an unvetted internet search or a general chatbot, not a clinician [2]. Safeguards commenters would credit include calibrated uncertainty, communication tested across health-literacy levels, working escalation, traceable sources and a route to a human [3].

Others stress added risk: one notes that patients may describe symptoms imprecisely, lack a clinician to check against, or use a tool instead of seeking care [4], and some back raising the risk of patient-facing outputs on triage, dosing or diagnosis [5]; one wants a persistent disclaimer that the tool doesn't replace a physician [6]. Some would replace the patient-versus-clinician split with whether users can verify outputs [7], demonstrated device-specific competency [8], a separate user dimension [9] or a tier for supervised non-clinical workers [10]. Of the two opposing points, one rejects treating credentials as proof of the ability to evaluate outputs [11], and the other calls patient empowerment a terrible goal [12].

Sources. [1] 0006 ¶p021; 0013 ¶p032; 0028 ¶p009; 0030 ¶p005; 0035 ¶p051; 0045 ¶p026; 0049 ¶p032; 0051 ¶p012; 0052 ¶p048; 0054 ¶p057; 0076 ¶p065; 0100 ¶p028; 0113 ¶p013; 0115 ¶p015; 0120 ¶p141 [2] 0006 ¶p021; 0021 ¶p024; 0100 ¶p028 [3] 0006 ¶p021; 0013 ¶p034; 0028 ¶p022; 0038 ¶p025; 0045 ¶p027; 0049 ¶p033; 0051 ¶p012; 0055 ¶p036, ¶p039; 0076 ¶p065; 0093 ¶p012; 0100 ¶p032, ¶p033; 0113 ¶p013 [4] 0118 ¶p014, ¶p015 [5] 0093 ¶p012; 0120 ¶p140 [6] 0118 ¶p016 [7] 0013 ¶p034 [8] 0037 ¶p006 [9] 0126 ¶p097 [10] 0009 ¶p014 [11] 0037 ¶p008 [12] 0043 ¶p009

Q4: Generalist vs specialist clinician users

18 comments, 23 tags: 22 conditional, 1 oppose

Several commenters treat professional title as only a proxy: what matters is whether the intended user can independently judge the output in that domain [1]. One cites a study in which incorrect suggestions, presented as AI output, sharply cut inexperienced readers' accuracy on mammograms [2]. Several want the expected user's competency stated and the device validated with that user group [3]. Labeling alone, some say, isn't enough: they would credit verified roles, access controls or review built into the workflow [4]. Another would let sponsors restrict users by specific training to lower the risk tier [5]. Several want devices to recognize, and be tested on recognizing, when specialist input is needed [6], while valuing wider access to specialist knowledge [7]. The one opposing point would replace the distinction with intended use and the qualifications it requires [8].

Sources. [1] 0005 ¶p252, ¶p305; 0028 ¶p022; 0049 ¶p034; 0052 ¶p050; 0055 ¶p044; 0056 ¶p006; 0113 ¶p014 [2] 0126 ¶p091, ¶p092 [3] 0003 ¶p025; 0051 ¶p013; 0055 ¶p046; 0118 ¶p018 [4] 0013 ¶p036; 0049 ¶p035; 0055 ¶p047, ¶p052 [5] 0129 ¶p008 [6] 0025 ¶p325; 0051 ¶p013; 0056 ¶p009; 0093 ¶p013 [7] 0025 ¶p322; 0049 ¶p034; 0051 ¶p013 [8] 0113 ¶p014

Q5: Multi-turn conversational risk

33 comments, 56 tags: 5 support, 51 conditional

Most commenters agree that risk must be judged across whole conversations, not single turns; all 56 points accept FDA's direction, 51 of them with additions or changes [1]. One found models worsening the longer its testers talked to them, and another identified individually accurate answers that can add up to undue reassurance [2].

Several propose trajectory tests: simulated users who minimize symptoms, resist escalation or press for advice; long conversations, new sessions and stored memory; and key facts withheld [3]. Two want results reported apart, such as timely action versus later recovery, or accuracy versus confident presentation [4].

Several would state intended use as a declared envelope of permitted behaviors or transitions [5], and several would judge a device by where its conversations can lead, such as the riskiest reachable output, not their opening turns [6]. One would instead use the highest level the device is designed to reach, and another says an out-of-scope capability should not by itself redefine intended use [7].

Some want devices to detect drift into directive or out-of-scope output and respond, such as by warning, escalating to a human or rechecking [8], and some want boundaries held by enforceable controls, such as deterministic filters outside the model or gates [9]; one wants them shown to hold with instructions absent or contradicted [10]. That commenter also recorded an assistant turning directive after inferring a user's frustration, and wants tests that vary only expressed emotion [11].

Sources. [1] 0003 ¶p026; 0005 ¶p306; 0006 ¶p022; 0013 ¶p038; 0017 ¶p023; 0020 ¶p014; 0025 ¶p100; 0028 ¶p010; 0035 ¶p048; 0045 ¶p031; 0049 ¶p036; 0055 ¶p053; 0065 ¶p034; 0076 ¶p025; 0093 ¶p014; 0105 ¶p022; 0113 ¶p015; 0115 ¶p016; 0118 ¶p020; 0120 ¶p149; 0127 ¶p021 [2] 0021 ¶p013; 0098 ¶p031 [3] 0006 ¶p022; 0013 ¶p038; 0021 ¶p014; 0025 ¶p109, ¶p110; 0045 ¶p031; 0055 ¶p054, ¶p057, ¶p058, ¶p059; 0065 ¶p039; 0105 ¶p024; 0111 ¶p012; 0115 ¶p016 [4] 0091 ¶p041; 0065 ¶p038 [5] 0003 ¶p027; 0006 ¶p022; 0035 ¶p049; 0051 ¶p014; 0093 ¶p014 [6] 0003 ¶p027; 0045 ¶p031; 0052 ¶p053; 0065 ¶p034; 0118 ¶p020; 0126 ¶p107 [7] 0131 ¶p041; 0120 ¶p150 [8] 0043 ¶p013; 0049 ¶p037; 0105 ¶p023; 0113 ¶p015 [9] 0052 ¶p054; 0055 ¶p061, ¶p070; 0120 ¶p151; 0127 ¶p031 [10] 0065 ¶p042 [11] 0065 ¶p016, ¶p022, ¶p035

Q6: Care escalation: under- vs over-escalation

35 comments, 49 tags: 1 support, 47 conditional, 1 oppose

Most commenters want under- and over-escalation measured and reported separately, not blended into one accuracy score [1]. Of the 49 points, 47 accept FDA's approach with changes and one as it stands; the one opposing point, rejecting AI in this area, says no trade-off is acceptable [2].

Many want sponsors to state and justify their trade-off for the clinical context, often in advance [3]; some reject one universal threshold or weighting [4].

Several would weight missed escalation more heavily, at least for serious conditions, some with near-zero tolerance or a separate pass-fail limit for severe misses [5]. Several also stress over-escalation's real costs, such as lost trust or police at the door [6]; one argues it can be the more dangerous error [7], and another that its diffuse harms are under-detected by postmarket monitoring [8].

Several say escalation is not just yes or no: it can be done badly, reach the wrong level, timing or destination, come in the wrong protocol order, break down at the handoff, or go through an intermediate

channel that makes false alarms cheaper [9]. One sees the two errors as sensitivity and specificity for each condition, not incommensurable [10]; another would set error bounds against disagreement among human raters [11]; and one says detailed weighting is often possible only in product-specific risk management [12].

Sources. [1] 0005 ¶p308; 0006 ¶p023; 0009 ¶p021; 0017 ¶p023; 0035 ¶p051; 0045 ¶p033; 0049 ¶p039; 0051 ¶p015; 0054 ¶p004; 0067 ¶p014; 0091 ¶p044; 0093 ¶p015; 0100 ¶p037; 0111 ¶p016; 0113 ¶p016; 0115 ¶p016; 0116 ¶p052; 0118 ¶p022; 0120 ¶p153 [2] 0020 ¶p016; 0019 ¶p016 [3] 0006 ¶p023; 0009 ¶p021; 0013 ¶p041; 0026 ¶p003; 0028 ¶p011; 0045 ¶p033; 0049 ¶p039; 0051 ¶p015; 0067 ¶p015; 0076 ¶p061; 0084 ¶p003; 0091 ¶p047; 0093 ¶p015; 0100 ¶p036; 0111 ¶p016; 0126 ¶p108 [4] 0009 ¶p021; 0028 ¶p011; 0084 ¶p003; 0093 ¶p015 [5] 0003 ¶p028; 0005 ¶p308; 0006 ¶p023; 0026 ¶p002; 0055 ¶p083; 0076 ¶p061; 0100 ¶p037; 0118 ¶p022; 0120 ¶p155 [6] 0003 ¶p028; 0006 ¶p023; 0021 ¶p015; 0025 ¶p117; 0084 ¶p002; 0111 ¶p015 [7] 0043 ¶p015 [8] 0013 ¶p043 [9] 0009 ¶p020; 0021 ¶p015; 0026 ¶p003; 0052 ¶p067; 0055 ¶p076; 0082 ¶p027 [10] 0126 ¶p108 [11] 0100 ¶p038 [12] 0052 ¶p060

PREMARKET: Evaluating a device before it is sold

77 comments, 751 tags: 65 support, 678 conditional, 6 oppose, 2 neutral

Q7: Competency-based approach overall

46 comments, 95 tags: 31 support, 64 conditional

Most commenters back the competency-based approach of benchmarking followed by clinical confirmation, and none of the 95 points opposes it: 31 support it as proposed and 64 ask for changes [1]. Several see it as the practical answer to open-ended, variable behavior that exhaustive or conventional testing can't cover [2].

Many endorse evaluating the deployed device, including parts such as prompts, retrieval and interface, rather than the foundation model alone [3]. Several want results bound to one exactly identified configuration, some with re-testing when it changes [4]. Two extend the device to a whole physical pathway, such as sensors and signal processing or a production line [5].

Many find the analogy to clinician training and credentialing apt [6], but several mark its limits: devices lack clinicians' accountability, can be silently swapped, fail at scale, or lack proven psychometrics [7]. Some say it therefore justifies the shape of the evidence, not lighter testing, looser change control or shared accountability [8]; one asks for a fast way to suspend a device's authorization [9].

Several caution that passing competency tests does not show clinical benefit or full safety, warning against a surrogate endpoint or wanting clinical validity kept apart from clinical utility [10]. Several would carry it past clearance, as part of ongoing oversight rather than a one-time exam, such as a longer licensure-like pipeline, a supervised stage with lighter entry evidence, or scope widened only through prespecified evidence gates with rollback [11].

Some want competency defined more fully, per clinical task or envelope, including recognizing uncertainty and knowing when not to act [12]. Three want it complemented, for example by assurance that evidence was current and actions authorized, by controls such as cybersecurity and version control, or by establishing clinical association before benchmarking [13]. One would make bounding inputs and outputs the primary expectation, with sponsors justifying why it isn't achievable before using competency-based evaluation [14]. Single commenters add that assessments should include pressure, hierarchy and unfamiliar patients [15]; that inference is arguably least reliable for rare must-not-miss conditions [16]; that training data must include real patient conversations [17]; and that sponsors should disclose whether users can see output before verification finishes [18].

Sources. [1] 0003 ¶p030; 0006 ¶p029; 0012 ¶p003; 0013 ¶p046; 0017 ¶p003; 0025 ¶p145; 0028 ¶p023; 0035 ¶p053; 0043 ¶p016; 0045 ¶p116; 0049 ¶p041; 0052 ¶p069; 0054 ¶p066; 0055 ¶p084; 0056 ¶p031; 0058 ¶p002; 0087 ¶p022; 0088 ¶p021; 0093 ¶p016; 0094 ¶p041; 0100 ¶p040; 0102 ¶p005; 0113 ¶p019; 0114 ¶p003; 0115 ¶p018; 0117 ¶p004; 0118 ¶p024; 0120 ¶p157; 0123 ¶p014; 0124 ¶p002 [2] 0006 ¶p029; 0028 ¶p023; 0052 ¶p069; 0054 ¶p066; 0093 ¶p016; 0102 ¶p005; 0114 ¶p003; 0117 ¶p004; 0123 ¶p014; 0124 ¶p002 [3] 0017 ¶p025; 0027 ¶p019; 0049 ¶p041; 0053 ¶p008; 0054 ¶p067; 0055 ¶p085; 0076 ¶p026; 0087 ¶p018; 0088 ¶p024; 0093 ¶p016; 0100 ¶p040; 0114 ¶p007; 0120 ¶p158; 0127 ¶p019 [4] 0005 ¶p309; 0035 ¶p053; 0076 ¶p047; 0088 ¶p024; 0115 ¶p018 [5] 0053 ¶p008; 0074 ¶p016 [6] 0003 ¶p011; 0006 ¶p029; 0046 ¶p017; 0052 ¶p072; 0058 ¶p053; 0076 ¶p046; 0102 ¶p005; 0103 ¶p014; 0106 ¶p013; 0117 ¶p004; 0118 ¶p024; 0123 ¶p014 [7] 0013 ¶p047, ¶p048; 0020 ¶p032, ¶p036; 0052 ¶p072; 0076 ¶p047; 0123 ¶p017, ¶p020 [8] 0013 ¶p051; 0020 ¶p036; 0076 ¶p051 [9] 0076 ¶p048 [10] 0020 ¶p030, ¶p071; 0049 ¶p042; 0055 ¶p086; 0056 ¶p004; 0094 ¶p019; 0118 ¶p026; 0126 ¶p124 [11] 0045 ¶p036; 0052 ¶p076, ¶p079; 0115 ¶p018; 0118 ¶p026; 0120 ¶p056, ¶p057 [12] 0025 ¶p147; 0056 ¶p004; 0114 ¶p006; 0123 ¶p019 [13] 0049 ¶p014; 0123 ¶p016; 0113 ¶p019 [14] 0126 ¶p139 [15] 0058 ¶p053 [16] 0126 ¶p127 [17] 0118 ¶p024 [18] 0098 ¶p041

Q8: Using the risk framework to scale premarket evidence

25 comments, 38 tags: 3 support, 34 conditional, 1 oppose

Most commenters agree that evidence should rise with a function's risk; of the 38 points, three support FDA's idea as proposed and 34 add conditions [1]. The one opposing point would let hazard analysis and risk-control effectiveness, not the two axes, set the evidence [2].

Several want FDA to publish a map from risk position to expected evidence, some as illustrative or non-binding, saying sponsors otherwise cannot plan [3].

Several sketch tiers, from benchmarking alone or light confirmation for low-risk informational functions up to prospective study or comparable rigor for the riskiest [4]. Two warn that demanding trials broadly would favor incumbents [5], while one wants prospective scrutiny generally for action-directing or action-taking devices, whatever the consequence level [6].

Some would set evidence by the risk left after safeguards, crediting controls or bounding [7]; one wants any such safeguard, clinician review for example, shown to work [8], and another wants the worst factor, not an average position, to govern [9]. One wants evidence scaled continuously, not in rigid bins [10], and another that risk position means little before clinical association is shown [11].

Sources. [1] 0020 ¶p063; 0025 ¶p333; 0035 ¶p116, ¶p120; 0045 ¶p039; 0049 ¶p044; 0051 ¶p016; 0052 ¶p071; 0056 ¶p032; 0067 ¶p017; 0072 ¶p007; 0091 ¶p060; 0100 ¶p040; 0113 ¶p035; 0118 ¶p028; 0120 ¶p167; 0123 ¶p003; 0126 ¶p161 [2] 0055 ¶p107 [3] 0003 ¶p031, ¶p032; 0006 ¶p024; 0013 ¶p053, ¶p055; 0051 ¶p016; 0093 ¶p017 [4] 0006 ¶p024; 0020 ¶p063; 0045 ¶p039; 0094 ¶p042; 0113 ¶p035; 0118 ¶p028 [5] 0006 ¶p024; 0094 ¶p024 [6] 0107 ¶p016 [7] 0072 ¶p007; 0120 ¶p164; 0126 ¶p133 [8] 0052 ¶p070 [9] 0005 ¶p310 [10] 0049 ¶p045 [11] 0113 ¶p021

Q9: Benchmarking structure and methods

57 comments, 203 tags: 17 support, 186 conditional

Many commenters find the benchmarking elements sound or a useful foundation, and no point opposes the approach: 17 support it as proposed and 186 ask for changes [1].

A recurring concern is that averages hide the failures that matter. Several want non-compensable floors or separate limits for safety-critical failures [2], and several want results shown across subgroups, settings and languages, with suggestions such as pediatric age bands, shifts in disease mix, and safety and usability for each group rather than statistical parity [3].

Proposed additions vary widely. Several want security threats addressed more fully, such as indirect prompt injection through documents or retrieved content, poisoned sources and supply-chain compromise, some as a dedicated element [4]; some want, or ask FDA to consider, tests of safe behavior when networks, cloud services or model APIs fail [5]. Several want provenance and faithfulness assessed in their own right: accurate citations, faithful patient-specific facts, no invented specifics, and claims checked against the record or source [6]. Three want benchmarking to test whether input was received at all [7]; two report a transcriber producing fluent text over silent audio [8]. Some want deference to users tested, such as yielding to claimed seniority or expert assertion, or endorsing a patient's or caregiver's risky plan [9].

On methods, several want behavior tested over whole conversations and long sessions, not single turns or fresh runs [10]. Some want reproducibility measured across repeated runs on decision-relevant content, or results reported as distributions, not assumed from settings such as zero temperature [11], and some want the tested configuration fully defined and recorded [12]. Several address calibration and deferral, suggesting confidence graded against the quality of the evidence, out-of-domain test sets that also count over-deferral, deferral on unverifiable claims scored on its own, and no credit for a model's

own statement of uncertainty [13]. Several want human factors and automation bias tested more directly, one linking communication quality to existing usability standards [14].

Some back structurally independent adjudicators, including language models [15], and three want an LLM judge's relationship to the device examined, such as a different model family, disclosed lineage or shown agreement with human panels [16]. Several would examine or credit safeguards built into the architecture, such as deterministic logic outside the model, some through design verification plus targeted testing [17]. Standards suggested include IEC 81001-5-1, IEC 62366-1, ISO/IEC 42001, ISO/IEC 23894, the NIST AI RMF and IMDRF's clinical evaluation guidance [18]; one asks FDA itself to say which standards are mature [19].

Domain-specific requests include specialty modules built with professional societies, spotting clinical need behind administrative requests, case-locked history-taking question lists, chronic disease reasoning, atypical presentations, image-grounded radiology reports and calibrated distress for exposure therapy [20]. Single commenters add that a fine-tuned model needs its own subgroup testing [21], that benchmarks should cover reasonably foreseeable use, such as questions about suicide [22], that a device describing its own safety limits does not show it keeps them [23], and that the strength of an output's wording should match the strength of the evidence behind it [24].

Sources. [1] 0003 ¶p033; 0005 ¶p311; 0013 ¶p056; 0020 ¶p015; 0025 ¶p154; 0028 ¶p024; 0043 ¶p018; 0045 ¶p042; 0049 ¶p047; 0052 ¶p081; 0056 ¶p010; 0067 ¶p018; 0078 ¶p004; 0091 ¶p012; 0093 ¶p018; 0094 ¶p044; 0100 ¶p041; 0114 ¶p008; 0118 ¶p030; 0120 ¶p169 [2] 0005 ¶p296; 0021 ¶p010; 0025 ¶p159; 0052 ¶p081; 0055 ¶p087; 0056 ¶p014; 0114 ¶p013 [3] 0017 ¶p053; 0020 ¶p060; 0028 ¶p024; 0047 ¶p003; 0051 ¶p017; 0058 ¶p052; 0059 ¶p029; 0118 ¶p031; 0126 ¶p292 [4] 0013 ¶p057; 0018 ¶p012; 0032 ¶p053; 0078 ¶p051; 0102 ¶p008; 0117 ¶p007 [5] 0032 ¶p046; 0098 ¶p030; 0111 ¶p050 [6] 0013 ¶p058; 0054 ¶p068; 0093 ¶p018; 0111 ¶p024; 0120 ¶p174; 0126 ¶p106 [7] 0062 ¶p031; 0064 ¶p038; 0078 ¶p034 [8] 0062 ¶p027; 0078 ¶p036 [9] 0058 ¶p025; 0098 ¶p034; 0123 ¶p023, ¶p026 [10] 0027 ¶p058; 0056 ¶p013; 0078 ¶p032; 0088 ¶p022; 0098 ¶p032; 0114 ¶p011 [11] 0076 ¶p013; 0088 ¶p032; 0126 ¶p152, ¶p205 [12] 0035 ¶p057; 0049 ¶p048; 0067 ¶p021 [13] 0026 ¶p004, ¶p006; 0059 ¶p002; 0076 ¶p050; 0098 ¶p039; 0127 ¶p033 [14] 0009 ¶p016; 0018 ¶p017; 0024 ¶p007; 0056 ¶p013; 0067 ¶p023; 0078 ¶p053; 0093 ¶p012; 0094 ¶p045 [15] 0076 ¶p027; 0126 ¶p118; 0127 ¶p022 [16] 0076 ¶p056; 0098 ¶p037; 0123 ¶p052 [17] 0035 ¶p055; 0052 ¶p083; 0100 ¶p042; 0111 ¶p021; 0127 ¶p031 [18] 0018 ¶p013; 0067 ¶p022; 0093 ¶p018; 0113 ¶p023 [19] 0013 ¶p061 [20] 0003 ¶p034; 0025 ¶p155; 0088 ¶p028; 0118 ¶p030, ¶p032; 0120 ¶p170; 0129 ¶p010 [21] 0027 ¶p063 [22] 0021 ¶p009 [23] 0065 ¶p028 [24] 0094 ¶p044

Q10: Benchmark validity and sponsor-developed benchmarks

37 comments, 74 tags: 74 conditional

All 74 points accept FDA's approach with changes. Many commenters agree that public benchmarks cannot stand alone, given contamination, saturation and limited representativeness [1]. Many want sponsors to show, not assume, that benchmark results predict real-world or clinical performance for the intended use [2]; one cites a study in which models alone identified conditions far more often than members of the public using them [3].

Most call for sequestered or held-out test sets, often held by an independent party such as a national repository, third-party custodian or professional society [4]. Contamination checks suggested include training-data cutoff attestations, dating items against that cutoff, canary items and freshly generated test sets [5], and one would treat public adversarial sets as potentially exposed [6]; one asks FDA to accept reasonable demonstrations of independence where upstream training data cannot be seen [7].

Many would permit sponsor-built benchmarks, often needed for narrow uses, with safeguards such as locking them in advance, keeping them out of development, independent review or adjudication, or a sequestered complement [8]. Two would not disfavor sponsor benchmarks for lacking independence: one prefers full disclosure, noting that small developers would otherwise have no benchmark [9], and the other documentation, sequestration and independent adjudication rather than exclusion [10].

Some treat the benchmark as a measuring instrument needing its own qualification: a reference standard fixed independently of the adjudicator, adjudicator agreement with humans by dimension, a documented reproducibility floor, a structured validity argument, or machine-checkable dataset documentation [11]. Some want items that reflect messy real patients, or premises that change the right answer [12]. One says a small, well-characterized dataset can be stronger evidence than a large one of unclear provenance [13].

Sources. [1] 0003 ¶p036; 0005 ¶p012; 0013 ¶p066; 0017 ¶p027; 0045 ¶p049; 0048 ¶p005; 0087 ¶p023; 0102 ¶p010; 0107 ¶p012; 0114 ¶p016; 0115 ¶p022; 0117 ¶p010; 0126 ¶p126 [2] 0005 ¶p312; 0013 ¶p064; 0017 ¶p027; 0028 ¶p024; 0035 ¶p060; 0048 ¶p005; 0049 ¶p049; 0051 ¶p001; 0093 ¶p019; 0100 ¶p044; 0113 ¶p024; 0115 ¶p022; 0116 ¶p049; 0118 ¶p034; 0123 ¶p040 [3] 0123 ¶p035 [4] 0003 ¶p037; 0005 ¶p254; 0013 ¶p065; 0017 ¶p027; 0035 ¶p060; 0045 ¶p049; 0049 ¶p050; 0051 ¶p018; 0056 ¶p014; 0067 ¶p025; 0088 ¶p016; 0093 ¶p019; 0100 ¶p046; 0102 ¶p010; 0114 ¶p017; 0115 ¶p022; 0116 ¶p048; 0117 ¶p010; 0118 ¶p034; 0120 ¶p179; 0123 ¶p042 [5] 0013 ¶p063; 0067 ¶p025; 0126 ¶p268 [6] 0027 ¶p073 [7] 0120 ¶p182 [8] 0003 ¶p037; 0013 ¶p065; 0028 ¶p024; 0035 ¶p060; 0048 ¶p005; 0049 ¶p050; 0051 ¶p018; 0056 ¶p014; 0067 ¶p025; 0093 ¶p019; 0113 ¶p024; 0118 ¶p034; 0120 ¶p181; 0123 ¶p042 [9] 0082 ¶p031 [10] 0100 ¶p047 [11] 0085 ¶p025; 0091 ¶p025, ¶p028; 0095 ¶p034; 0123 ¶p033 [12] 0025 ¶p172; 0082 ¶p029; 0102 ¶p010; 0117 ¶p010 [13] 0074 ¶p026

Q11: Choosing a clinical confirmation approach

31 comments, 46 tags: 4 support, 40 conditional, 2 oppose

Most commenters back a ladder of confirmation methods scaled to risk, so that a prospective study is not always required; of the 46 points, 4 support FDA's approach as proposed and 40 add conditions [1]. The two opposing points want a clinical trial in every case, or prospective studies as the main proof for devices that influence care [2].

Many would reserve prospective studies for, or require them in, autonomous, action-directing, patient-facing or high-consequence uses [3], and three warn that requiring them broadly would favor large incumbents [4]. Two want the required rung fixed by the risk assessment, one explicitly rather than by sponsor choice [5], while three tie it to the uncertainty left after benchmarking [6].

Some favor shadow deployment, one noting that curated retrospective data drop the messy inputs that cause failures [7]. Several want confirmation to reflect the populations, settings and inputs of real use, some with the expected input distribution defined in advance [8]. One wants reference calls recorded before reviewers see the output [9]; another, that retrospective tests withhold information unavailable at the decision point [10]; and one argues that strong benchmark results can justify lighter confirmation only as far as the benchmark itself is qualified [11].

Sources. [1] 0003 ¶p038; 0005 ¶p314; 0006 ¶p024; 0009 ¶p023; 0013 ¶p068; 0017 ¶p028; 0020 ¶p017; 0028 ¶p015; 0030 ¶p008; 0045 ¶p052; 0049 ¶p051; 0052 ¶p085; 0055 ¶p153; 0087 ¶p024; 0093 ¶p020; 0094 ¶p048; 0100 ¶p040; 0113 ¶p025; 0118 ¶p036; 0120 ¶p186 [2] 0019 ¶p028; 0107 ¶p007 [3] 0003 ¶p039; 0005 ¶p314; 0006 ¶p024; 0017 ¶p028; 0025 ¶p335; 0049 ¶p052; 0055 ¶p156; 0086 ¶p004; 0093 ¶p020; 0113 ¶p025; 0120 ¶p187 [4] 0003 ¶p039; 0006 ¶p024; 0094 ¶p048 [5] 0013 ¶p069; 0045 ¶p052 [6] 0030 ¶p009; 0052 ¶p085; 0055 ¶p153 [7] 0013 ¶p070; 0049 ¶p052; 0051 ¶p001; 0055 ¶p155; 0118 ¶p036 [8] 0005 ¶p255; 0013 ¶p071; 0055 ¶p157; 0093 ¶p020; 0113 ¶p025 [9] 0009 ¶p027 [10] 0052 ¶p086 [11] 0091 ¶p061

V-RWD: Real-world and ex-US data in clinical confirmation (no numbered question)

3 comments, 4 tags: 4 conditional

All four points accept real-world and non-US data in clinical confirmation, with conditions [1]. One wants a formal transportability assessment, with adjustment where settings differ [2]; another warns that such data leave thinly sampled combinations thin and bring population, practice and equipment differences whose relevance must be shown [3]. The third wants independently governed longitudinal outcome databanks listed as a confirmation approach [4].

Sources. [1] 0051 ¶p019; 0094 ¶p047; 0126 ¶p261 [2] 0051 ¶p019 [3] 0126 ¶p261 [4] 0094 ¶p047

Q12: Statistical rigor and combining evidence

23 comments, 39 tags: 38 conditional, 1 neutral

All but one of the 39 points ask for changes; the neutral one suggests Qualtrics studies, noting consent concerns [1]. On combining evidence, many advise against pooling benchmark with confirmation results, or synthetic with real data, into one estimate, reporting them separately unless conditions such as a shared estimand, shown agreement or prespecified weighting are met [2]; one would let confirmation results govern acceptance [3]. By contrast, one asks that small sponsors be allowed a single prospective feasibility study serving both purposes [4].

Several stress that outputs vary for a fixed input, wanting repeated runs and their variability reported [5]. Suggestions include judging acceptance on a lower confidence bound, comparing differences with the device's own run-to-run floor, computing intervals per case rather than per run, and stating the failure-rate bound behind zero-failure results [6].

Several want endpoints, units of analysis and intervals prespecified, with results broken down by subgroup, site or severity [7], and some warn that averages and realistically sized samples say little about rare or thinly sampled, high-consequence cases [8]. One wants a minimum detectable difference stated for any null subgroup result [9].

Sources. [1] 0019 ¶p030 [2] 0005 ¶p300; 0013 ¶p077; 0049 ¶p053; 0051 ¶p020; 0055 ¶p170, ¶p171; 0091 ¶p065; 0093 ¶p021; 0120 ¶p192; 0126 ¶p273 [3] 0013 ¶p077 [4] 0129 ¶p012 [5] 0013 ¶p073; 0020 ¶p045; 0035 ¶p054; 0055 ¶p162; 0076 ¶p054; 0095 ¶p030; 0120 ¶p190; 0123 ¶p044 [6] 0013 ¶p074; 0095 ¶p030; 0123 ¶p043, ¶p048 [7] 0005 ¶p315; 0020 ¶p045; 0045 ¶p060; 0052 ¶p088; 0055 ¶p159, ¶p164; 0120 ¶p189; 0123 ¶p048; 0126 ¶p235 [8] 0020 ¶p046; 0035 ¶p054; 0052 ¶p087; 0126 ¶p260 [9] 0095 ¶p038

Q13: Suitability and safeguards for synthetic data

26 comments, 49 tags: 2 support, 46 conditional, 1 neutral

Of the 49 points, 46 accept a role for synthetic data with conditions, two agree with FDA as proposed and one is neutral; most commenters see it as a supplement to real data, not a replacement [1]. Many find it best suited to rare events, stress, adversarial or boundary tests, and controlled variation of single factors [2]. Several say it cannot by itself establish real-world performance rates, representativeness or subgroup performance [3], and some add human behavior, workflow or clinical outcomes, one warning that synthetic patients come out too tidy [4].

Many share FDA's concern that generators from the same model class can reproduce the device's blind spots [5]. Proposed safeguards include independent or differently derived generators, disclosure of the generator and method, validation against real distributions, confirmation on real or registry data, and separate reporting [6]; one adds that a generating model must not certify its own labels [7], and another knows of no validated way to show a synthetic population matches a real one [8]. One commenter is more optimistic, calling validated synthetic data, such as physics-based imaging models, the most direct route to stronger generalizability evidence [9], and another would build synthetic subpopulations for non-rare conditions from meta-analyses rather than single studies [10].

Sources. [1] 0003 ¶p042; 0005 ¶p316; 0017 ¶p033; 0020 ¶p050; 0025 ¶p348; 0035 ¶p062; 0045 ¶p062; 0048 ¶p006; 0049 ¶p056; 0052 ¶p089; 0055 ¶p179; 0091 ¶p063; 0113 ¶p027; 0115 ¶p023; 0120 ¶p195; 0126 ¶p276 [2] 0005 ¶p316; 0013 ¶p081; 0017 ¶p033; 0025 ¶p339; 0045 ¶p063; 0048 ¶p006; 0049 ¶p055; 0052 ¶p089; 0055 ¶p172; 0091 ¶p063; 0093 ¶p022; 0113 ¶p027; 0115 ¶p023; 0120 ¶p194; 0126 ¶p264 [3] 0003 ¶p042; 0013 ¶p082, ¶p083; 0048 ¶p006; 0051 ¶p001; 0076 ¶p055; 0091 ¶p064; 0093 ¶p022; 0113 ¶p027; 0118 ¶p040 [4] 0025 ¶p340; 0055 ¶p180, ¶p182, ¶p185; 0126 ¶p271 [5] 0003 ¶p041; 0005 ¶p260; 0013 ¶p079; 0017 ¶p033;

0020 ¶p048; 0028 ¶p016; 0045 ¶p063; 0049 ¶p056; 0055 ¶p186; 0076 ¶p055; 0093 ¶p022; 0113 ¶p027; 0118 ¶p040; 0120 ¶p196; 0126 ¶p274 [6] 0005 ¶p260; 0013 ¶p084; 0017 ¶p033; 0028 ¶p016; 0045 ¶p063; 0049 ¶p056; 0055 ¶p187; 0076 ¶p055; 0085 ¶p038; 0093 ¶p022; 0115 ¶p023; 0118 ¶p040; 0120 ¶p197; 0126 ¶p275 [7] 0091 ¶p066 [8] 0013 ¶p085 [9] 0126 ¶p258, ¶p275 [10] 0043 ¶p022

Q14: Comparators and acceptance criteria for open-ended outputs

33 comments, 85 tags: 1 support, 83 conditional, 1 oppose

Of the 85 points, 83 accept FDA's approach with changes, one agrees that no single comparator suits every device [1], and one doubts that clinician-panel comparators make sense, the commenter knowing few physicians who would join one [2].

Most commenters want the human-AI team evaluated, not just the device [3]. Many would make team performance the main basis when a clinician reviews the output [4]; several still want device-alone results [5], and some keep device-alone comparison for autonomous use [6]. Several want team evaluation to measure automation bias [7], and three say human oversight should not be assumed to work as a safeguard [8]. One argues that team performance is undefined until the output's place in the workflow is fixed, citing an experiment in which the timing of AI support alone changed lay participants' trust [9]; another would stop sponsors testing as a team while marketing for autonomous use [10].

Commenters differ on the clinician standard. Several oppose or caution against the median clinician in practice as the main standard, generally or for specialty tasks, saying it normalizes existing gaps in care, and preferring specialists, the standard of care, an evidence-based standard or consequence-based criteria [11]. Others accept a median clinician: as the default, with consensus panels as a ceiling [12]; where no ground truth exists [13]; as extra comparison data beside a standard-of-care floor [14]; or, where no specialist is available, a median specialist used to set a fixed threshold [15]. Many want specialists to set or judge the reference for specialist or high-stakes tasks [16], while some would match the comparator to the intended user or task [17].

Several want clinician panels' own agreement measured and reported [18]. Several want acceptance criteria or comparators prespecified [19], and some want criteria to cover errors such as omissions and unjustified certainty, not just the final answer [20]. One proposes conjunctive standards, so that safety failures cannot be offset by strength elsewhere [21]; another, for higher-risk functions, structured clinical claims scored against a reference standard [22].

Sources. [1] 0126 ¶p174 [2] 0019 ¶p034 [3] 0005 ¶p317; 0013 ¶p094; 0017 ¶p004; 0025 ¶p198; 0028 ¶p017; 0035 ¶p062; 0046 ¶p004; 0052 ¶p094; 0054 ¶p006; 0055 ¶p192; 0056 ¶p016; 0058 ¶p036; 0070 ¶p009; 0076 ¶p014; 0086 ¶p005; 0087 ¶p027; 0093 ¶p023; 0094 ¶p050; 0100 ¶p051; 0118 ¶p042; 0120 ¶p202; 0123 ¶p053 [4] 0013 ¶p094; 0028 ¶p017; 0035 ¶p062; 0052 ¶p094; 0056 ¶p017; 0070 ¶p009; 0093 ¶p023; 0094 ¶p050; 0100 ¶p051; 0118 ¶p042; 0120 ¶p202; 0123 ¶p053 [5] 0005 ¶p317; 0013 ¶p094; 0028 ¶p024; 0046 ¶p004; 0055 ¶p192; 0056 ¶p017; 0120 ¶p203; 0123 ¶p053 [6] 0035 ¶p062; 0093 ¶p023; 0094 ¶p050; 0100 ¶p051; 0118 ¶p042 [7] 0013 ¶p095; 0017 ¶p031; 0054 ¶p006; 0076 ¶p059; 0086 ¶p005; 0120 ¶p092; 0123 ¶p053 [8] 0054 ¶p076; 0056 ¶p017; 0058 ¶p035 [9] 0070 ¶p012, ¶p013 [10] 0094 ¶p051 [11] 0003 ¶p043, ¶p044; 0013 ¶p088, ¶p090; 0025 ¶p184, ¶p185; 0056 ¶p015; 0076 ¶p059 [12] 0100 ¶p049 [13] 0113 ¶p029 [14] 0055 ¶p191 [15] 0126 ¶p180, ¶p181 [16] 0003 ¶p044; 0055 ¶p190; 0056 ¶p009; 0076 ¶p059; 0093 ¶p023; 0113 ¶p029; 0118 ¶p043; 0120 ¶p199; 0123 ¶p050 [17] 0006 ¶p025; 0013 ¶p093; 0045 ¶p029; 0049 ¶p057 [18] 0009 ¶p033; 0013 ¶p092; 0043 ¶p023; 0100 ¶p049; 0106 ¶p013; 0123 ¶p049 [19] 0013 ¶p090; 0028 ¶p024; 0056 ¶p018; 0093 ¶p023; 0113 ¶p029 [20] 0049 ¶p058; 0055 ¶p189; 0056 ¶p018; 0120 ¶p201 [21] 0123 ¶p051 [22] 0126 ¶p151

Q15: Comparing with care in the device's absence

24 comments, 34 tags: 2 support, 31 conditional, 1 oppose

Of the 34 points, 31 accept FDA's idea with changes, two agree as proposed, and one opposes it, the commenter finding the question not yet answerable and asking who could justify such a comparison for AI [1]. Most commenters accept comparing a device with likely care in its absence, at least for some

purposes [2], several stressing devices that widen access, where the real alternative may be delay, internet searching or nothing [3]. Several want that alternative shown with data, such as care-access, utilization or turnaround records, rather than assumed [4]. Several would not let it lower the safety bar, keeping an absolute safety floor or standard alongside it [5].

One wants the comparator carried into labeling to prevent unevaluated scope creep [6]; another, writing on communication aids for nonverbal users, would add a pass-or-fail test in which any output from input carrying no signal fails [7]; a third considers no intervention the only reliable comparator [8].

Sources. [1] 0019 ¶p036, ¶p037 [2] 0003 ¶p044; 0005 ¶p318; 0006 ¶p025; 0009 ¶p035; 0013 ¶p096; 0017 ¶p032; 0020 ¶p041; 0025 ¶p200; 0045 ¶p065; 0049 ¶p060; 0051 ¶p021; 0052 ¶p092; 0055 ¶p193; 0064 ¶p009; 0082 ¶p008; 0093 ¶p024; 0100 ¶p050; 0113 ¶p030; 0118 ¶p046; 0120 ¶p205; 0123 ¶p050; 0126 ¶p182 [3] 0003 ¶p044; 0006 ¶p025; 0013 ¶p096; 0049 ¶p060; 0100 ¶p050; 0118 ¶p046; 0120 ¶p209 [4] 0006 ¶p025; 0009 ¶p040; 0013 ¶p097; 0051 ¶p021; 0093 ¶p024; 0100 ¶p050; 0120 ¶p210 [5] 0017 ¶p032; 0025 ¶p203; 0118 ¶p046; 0123 ¶p050; 0126 ¶p183 [6] 0013 ¶p097 [7] 0064 ¶p030, ¶p037 [8] 0043 ¶p024

Q16: Role of independent third parties

32 comments, 60 tags: 5 support, 54 conditional, 1 oppose

Of the 60 points, 54 accept third-party roles with conditions, five support them as proposed, and one opposes a third-party requirement, warning that such organizations could game the system and profit from drawn-out testing [1]. Most commenters would have third parties hold sequestered or hidden test data, adjudicate results, or both [2]. Three see them as necessary: for premarket testing generally, for the safety-critical recognition and escalation elements, or for medium- and high-risk devices [3]. Several would keep third-party review optional or let sponsors show equivalent evidence another way [4].

Most also want safeguards so third parties do not limit competition, such as several qualified bodies, published criteria, and no exclusive or mandatory gate [5]; some want fees published, capped, scaled or subsidized so small sponsors are not priced out [6]. Many point to ASCA or MDDT as models [7].

Many set independence conditions, such as disclosed or no financial ties and pay that does not depend on results [8], some extending them to the foundation model developer [9] and two to AI adjudicators sharing the device model's lineage [10]; one would define independence relative to the device under evaluation, so device developers can still build evaluation tools [11]. Several say third parties should not take over FDA's decisions or the manufacturer's accountability [12]. Some name professional societies among suitable third parties [13]; three prefer qualifying evaluation tools or open methods to certifying devices [14]; and one warns that, as far as the commenter knows, the needed capacity does not yet exist at scale [15].

Sources. [1] 0043 ¶p025 [2] 0005 ¶p319; 0006 ¶p028; 0013 ¶p098; 0025 ¶p210; 0028 ¶p017; 0035 ¶p064; 0045 ¶p069; 0049 ¶p062; 0055 ¶p195; 0056 ¶p019; 0076 ¶p057; 0091 ¶p067; 0093 ¶p025; 0094 ¶p055; 0100 ¶p052; 0115 ¶p024; 0118 ¶p048; 0120 ¶p214; 0123 ¶p042 [3] 0047 ¶p005; 0071 ¶p012; 0113 ¶p031 [4] 0013 ¶p103; 0094 ¶p055; 0100 ¶p054; 0120 ¶p213; 0126 ¶p291 [5] 0003 ¶p046; 0005 ¶p319; 0006 ¶p028; 0013 ¶p100; 0025 ¶p211; 0028 ¶p017; 0035 ¶p064; 0045 ¶p069; 0049 ¶p063; 0055 ¶p203; 0056 ¶p021; 0071 ¶p015; 0076 ¶p057; 0091 ¶p068; 0093 ¶p025; 0094 ¶p055; 0100 ¶p053; 0115 ¶p024; 0118 ¶p048; 0120 ¶p216; 0126 ¶p291 [6] 0003 ¶p045, ¶p046; 0006 ¶p028; 0094 ¶p055; 0100 ¶p053 [7] 0003 ¶p046; 0006 ¶p028; 0028 ¶p017; 0045 ¶p069; 0067 ¶p028; 0071 ¶p015; 0076 ¶p057; 0093 ¶p025; 0094 ¶p055; 0116 ¶p050; 0126 ¶p294 [8] 0005 ¶p319; 0013 ¶p102; 0017 ¶p034; 0025 ¶p211; 0035 ¶p064; 0045 ¶p069; 0049 ¶p063; 0055 ¶p202; 0056 ¶p020; 0071 ¶p014; 0093 ¶p025; 0100 ¶p055; 0113 ¶p031; 0115 ¶p024; 0118 ¶p048 [9] 0067 ¶p028; 0071 ¶p014; 0093 ¶p025; 0100 ¶p055; 0113 ¶p031 [10] 0071 ¶p014; 0100 ¶p055 [11] 0126 ¶p284 [12] 0045 ¶p121; 0049 ¶p063; 0055 ¶p204; 0093 ¶p025; 0116 ¶p050; 0120 ¶p217; 0126 ¶p291 [13] 0114 ¶p027; 0116 ¶p050; 0118 ¶p048; 0123 ¶p042 [14] 0067 ¶p028; 0091 ¶p067; 0126 ¶p291 [15] 0013 ¶p104

Q17: Other model architectures

16 comments, 24 tags: 24 conditional

All 24 points accept the approach for other architectures, with changes. Most commenters say it can extend to multimodal or world-model devices with added tests [1], and several want it, or see it as, architecture-neutral, judging what a device does rather than its model type [2]. For multimodal devices, proposed tests include cross-modal consistency [3], missing or mismatched images [4], and variation in equipment, image quality or skin tone [5]; for world models, error over predicted steps and bounded variability [6]. One reports an experiment in which models gave diagnoses with no image attached, and wants abstention measured per output channel [7]; another wants guidance specific to multimodal devices [8]. Others caution that thinner benchmarks should shift weight to clinical confirmation [9], that conversational tests do not fit imaging measurement functions [10], and that content-generating and world-model devices need different benchmark elements [11].

Sources. [1] 0005 ¶p321; 0049 ¶p064; 0055 ¶p108; 0093 ¶p026; 0108 ¶p001; 0113 ¶p033; 0118 ¶p050; 0120 ¶p219; 0123 ¶p054 [2] 0005 ¶p321; 0025 ¶p376; 0045 ¶p071; 0049 ¶p064; 0118 ¶p050 [3] 0005 ¶p321; 0049 ¶p064; 0055 ¶p109; 0113 ¶p033; 0120 ¶p220 [4] 0055 ¶p110; 0108 ¶p003; 0123 ¶p055 [5] 0055 ¶p113; 0093 ¶p026; 0118 ¶p050; 0123 ¶p057 [6] 0055 ¶p114; 0113 ¶p033 [7] 0108 ¶p002, ¶p003, ¶p007, ¶p010 [8] 0003 ¶p047 [9] 0013 ¶p106 [10] 0126 ¶p167 [11] 0129 ¶p013, ¶p014

POSTMARKET: Oversight once a device is in use

90 comments, 639 tags: 19 support, 610 conditional, 7 oppose, 3 neutral

Q18: Trading premarket uncertainty for postmarket monitoring

43 comments, 88 tags: 2 support, 79 conditional, 7 oppose

Of the 88 points, 79 accept relying more on postmarket monitoring under conditions, two agree with FDA's framing as it stands, and seven oppose the trade, holding that monitoring should verify or complement premarket evidence, not replace it [1].

Most commenters would rule out or limit reduced premarket evidence for functions that are high-consequence, irreversible, autonomous or time-critical, or whose failures could cause harm before monitoring catches them [2]. One would also exclude functions that produce permanent implants [3], and two, devices whose users cannot report errors or that run where monitoring cannot reach [4]. Some warn that the trade shifts risk onto patients [5].

Many want a prespecified monitoring plan with thresholds, triggers and set consequences such as restriction, suspension or rollback [6], two insisting that it be a binding condition of authorization [7]. Many also want evidence that the monitoring can actually detect the relevant failures, not just that a program exists [8]; some add that detection must come before harm [9], and one that sponsors state what their monitoring will not detect [10]. Some want monitoring records that the device or sponsor cannot rewrite [11], or automatic or immediate ways to stop the device [12], and two want monitoring in place from the first day of deployment [13]. Others suggest a staged rollout from shadow mode to wider use [14], want deploying sites able, among other things, to inventory, log and stop their devices [15], or suggest crediting machine supervision only once its own reliability is shown [16].

Some doubt that surveillance will see GenAI failures, citing passive under-reporting, harms no one specified in advance, postmarket duties met less reliably than premarket ones, and failures that never add up to a rate [17]. At the other end, some say real-world use yields evidence no premarket test can [18], and one would make the trade the default outside the highest-risk band [19].

Sources. [1] 0023 ¶p006; 0086 ¶p008; 0113 ¶p001, ¶p036; 0126 ¶p188 [2] 0005 ¶p322; 0013 ¶p114; 0017 ¶p036; 0020 ¶p055; 0024 ¶p009; 0025 ¶p217; 0028 ¶p025; 0035 ¶p066; 0045 ¶p075; 0048 ¶p007; 0049 ¶p068; 0051 ¶p022; 0052 ¶p119; 0053 ¶p009; 0054 ¶p080; 0055 ¶p214; 0075 ¶p021; 0076 ¶p035; 0077 ¶p050; 0090 ¶p015; 0093 ¶p027; 0094 ¶p061; 0100 ¶p057; 0105 ¶p026; 0115 ¶p025; 0116 ¶p025; 0118 ¶p052; 0120 ¶p228; 0126 ¶p194; 0131 ¶p027 [3] 0053 ¶p009 [4] 0077 ¶p043, ¶p048; 0093 ¶p027 [5] 0025 ¶p216; 0076 ¶p035; 0086 ¶p008; 0126 ¶p188 [6] 0003 ¶p050; 0006 ¶p026; 0013 ¶p111; 0028 ¶p025; 0035 ¶p066; 0055 ¶p212; 0075 ¶p020; 0090 ¶p014; 0094 ¶p061; 0099 ¶p025; 0100 ¶p057; 0120 ¶p116 [7] 0003 ¶p050; 0013 ¶p111 [8] 0005 ¶p322; 0008 ¶p013; 0013 ¶p112; 0033 ¶p013; 0035 ¶p066; 0049 ¶p067; 0077 ¶p031; 0090 ¶p015; 0093 ¶p027; 0099 ¶p024; 0116 ¶p011; 0126 ¶p190 [9] 0005 ¶p322; 0013 ¶p113; 0049 ¶p067; 0075 ¶p021; 0126 ¶p191 [10] 0033 ¶p014 [11] 0077 ¶p035; 0090 ¶p013; 0127 ¶p039; 0130 ¶p007 [12] 0102 ¶p006; 0117 ¶p006; 0126 ¶p192; 0130 ¶p008 [13] 0090 ¶p011; 0116 ¶p024 [14] 0054 ¶p080 [15] 0130 ¶p004, ¶p005, ¶p007, ¶p008 [16] 0111 ¶p038 [17] 0013 ¶p110; 0033 ¶p010; 0076 ¶p036; 0077 ¶p024 [18] 0003 ¶p049; 0006 ¶p026; 0100 ¶p057 [19] 0094 ¶p057

Q19: Postmarket performance evaluation approaches

71 comments, 191 tags: 11 support, 178 conditional, 2 neutral

Of the 191 points, 178 accept FDA's approach with changes, 11 support it as proposed and two are neutral. Many commenters call periodic re-benchmarking, sampled clinician review and degradation monitoring reasonable or useful [1], several stressing that the three are complements rather than alternatives [2]. One reports running such a program for years across millions of interactions [3]; another

found that public summaries of 95 authorized change control plans, mostly for non-generative devices, describe no scheduled re-benchmarking of unchanged models and no subgroup monitoring [4].

Many want reassessment both on a schedule and after triggering events [5]. Common triggers include changes to the foundation model, prompts, retrieval sources or tools, shifts in inputs or populations, new sites, and complaint or safety signals [6]. Two propose default intervals by risk tier, from monthly or quarterly checks for high-risk devices to annual review for low-risk ones [7]; one adds changes on the deploying site's side, such as a new EHR version [8].

Several want monitoring to look beyond model accuracy, at near misses, overrides, subgroup performance, human-AI interaction or patient outcomes [9], some wanting device use traced through to downstream care and outcomes [10]. Several want review samples to over-represent high-risk, unusual or disputed cases rather than only mirror routine use [11]; one wants reviewer agreement measured and reviewers calibrated [12]. Two suggest FDA clarify when a wrong generative output counts as a reportable malfunction, one proposing that errors users catch be reported in aggregate as rates [13].

Several want records detailed enough to reconstruct an event, such as the model version, configuration, inputs, outputs and human review behind an output [14]. Several want at least one check or record that does not rely on the device's account of itself [15], and several want monitoring to limit the patient data it keeps, for example through metadata, hashes or graduated access [16]. Several want each signal tied to a prespecified response or owner, such as restriction, rollback or suspension [17].

Some point to blind spots: real-world samples, unless supplemented with adversarial probes, do not test resistance to adversarial inputs [18]; a re-benchmark starts a fresh session and misses degradation that builds within one [19]; failures upstream of the model, such as dropped audio, may leave no record [20]; monitoring that waits for users to report errors fails users who cannot speak or check outputs [21]; and rates cannot be compared unless their populations are declared [22]. Proposed additions include sentinel or canary cases [23], real-time checking of each output against current sources [24], device-generated integrity metrics [25], re-examination on the original competencies, as in clinician re-certification [26], a pooled sentinel network across manufacturers [27], qualified data systems for higher-risk functions [28], and watching litigation and state law as outside signals [29]. Some ask that monitoring stay proportionate: continuous monitoring need not mean continuous submissions [30], obligations should not shut out small developers [31], and intensity should match a device's sources of variation [32].

Sources. [1] 0013 ¶p116; 0020 ¶p018; 0029 ¶p020; 0046 ¶p019; 0049 ¶p069; 0052 ¶p121; 0058 ¶p039; 0063 ¶p009; 0067 ¶p029; 0071 ¶p017; 0087 ¶p029; 0090 ¶p021; 0093 ¶p028; 0100 ¶p058; 0115 ¶p026; 0116 ¶p043; 0118 ¶p054; 0120 ¶p230; 0126 ¶p196 [2] 0013 ¶p116; 0052 ¶p121; 0071 ¶p017; 0090 ¶p021; 0093 ¶p028; 0118 ¶p054 [3] 0100 ¶p058 [4] 0090 ¶p018, ¶p019 [5] 0003 ¶p053; 0005 ¶p323; 0013 ¶p117; 0017 ¶p039; 0035 ¶p073; 0045 ¶p078; 0049 ¶p070; 0052 ¶p123; 0054 ¶p081; 0056 ¶p022; 0058 ¶p041; 0075 ¶p022; 0090 ¶p022; 0091 ¶p058; 0093 ¶p028; 0100 ¶p060; 0105 ¶p027; 0115 ¶p026; 0116 ¶p044; 0118 ¶p054 [6] 0003 ¶p053; 0005 ¶p323; 0013 ¶p117; 0035 ¶p074; 0049 ¶p070; 0052 ¶p123; 0054 ¶p092; 0056 ¶p023; 0105 ¶p028; 0115 ¶p027; 0120 ¶p234 [7] 0003 ¶p052; 0113 ¶p037 [8] 0130 ¶p010 [9] 0017 ¶p038; 0025 ¶p228; 0039 ¶p004; 0047 ¶p008; 0052 ¶p121; 0054 ¶p091; 0055 ¶p223; 0056 ¶p024; 0086 ¶p009; 0087 ¶p029; 0120 ¶p231 [10] 0021 ¶p018; 0025 ¶p224; 0051 ¶p023; 0116 ¶p013 [11] 0055 ¶p225; 0056 ¶p023; 0075 ¶p024; 0092 ¶p005; 0100 ¶p059; 0116 ¶p043; 0118 ¶p054 [12] 0033 ¶p024, ¶p026 [13] 0013 ¶p129; 0033 ¶p038 [14] 0025 ¶p256; 0035 ¶p114; 0042 ¶p059; 0047 ¶p006; 0049 ¶p071; 0050 ¶p009; 0052 ¶p141; 0053 ¶p013; 0075 ¶p052; 0087 ¶p043; 0092 ¶p015; 0098 ¶p043; 0102 ¶p009; 0105 ¶p041; 0116 ¶p031; 0117 ¶p008 [15] 0008 ¶p015; 0063 ¶p028; 0078 ¶p088; 0093 ¶p028; 0103 ¶p015; 0127 ¶p039 [16] 0035 ¶p067; 0042 ¶p032; 0049 ¶p074; 0075 ¶p050; 0101 ¶p008; 0116 ¶p036 [17] 0047 ¶p009; 0052 ¶p110; 0054 ¶p121; 0055 ¶p226; 0067 ¶p030; 0075 ¶p022; 0105 ¶p028; 0116 ¶p044; 0118 ¶p055 [18] 0027 ¶p042 [19] 0063 ¶p015 [20] 0062 ¶p044, ¶p046 [21] 0077 ¶p007 [22] 0131 ¶p043, ¶p045 [23] 0093 ¶p028; 0123 ¶p064 [24] 0071 ¶p020 [25] 0111 ¶p029 [26] 0058 ¶p041 [27] 0021 ¶p018 [28] 0116 ¶p022 [29] 0025 ¶p404 [30] 0087 ¶p030 [31] 0111 ¶p040 [32] 0126 ¶p207, ¶p208

Q20: Machine-based supervisory agents

43 comments, 91 tags: 2 support, 89 conditional

Of the 91 points, 89 accept supervisory agents with conditions and two support them as proposed, one noting that an autonomous breast-screening device certified in the EU already relies on such a layer [1]. Several see automated supervision as useful or as the only approach that scales to high volumes [2].

Most want the supervisor validated in its own right, for example by measuring its missed and false alarms against human adjudication and re-checking them over time [3]. Most also warn that a supervisor sharing the device's model, data or maker may share its blind spots, and want it independent of the device [4], though two caution that a different vendor or model does not by itself establish independence or reliability [5]. Several want the supervisor under version and change control [6].

Many would let supervisors triage or flag cases but keep final judgment with humans [7], three adding that humans should also sample cases the supervisor cleared [8]. Some see deterministic, rule-based checks as the right tool for fixed limits [9], and some want the supervisor's role and authority defined, from observing to blocking actions [10]; one suggests an observation-only phase first [11], while another doubts that a supervisor which can only observe adds safety [12].

Some report firsthand failures: a language model reviewing evidence made confident, unsupported claims that only the operator's own knowledge caught [13], and a verifier with too narrow a reference record would have flagged a true, clinically important statement [14]; another cites a study in which judge models' verdicts shifted under reworded prompts [15]. Others want supervisor output kept distinguishable from human judgment in the record [16], each finding to state the source consulted and its date [17], or a supervisor that may read every record but write nothing to the record of fact [18]; one suggests checking that records can express the supervisory relationship at all [19], and another would not let supervision justify less premarket evidence [20].

Sources. [1] 0126 ¶p210 [2] 0013 ¶p119; 0025 ¶p238; 0052 ¶p130; 0090 ¶p024; 0115 ¶p034; 0120 ¶p236 [3] 0003 ¶p055; 0005 ¶p324; 0013 ¶p123; 0017 ¶p040; 0025 ¶p240; 0028 ¶p019; 0033 ¶p030; 0035 ¶p076; 0045 ¶p081; 0049 ¶p079; 0052 ¶p130; 0055 ¶p229; 0066 ¶p047; 0071 ¶p024; 0075 ¶p044; 0076 ¶p063; 0090 ¶p025; 0093 ¶p029; 0100 ¶p062; 0102 ¶p010; 0106 ¶p017; 0111 ¶p035; 0113 ¶p038; 0115 ¶p037; 0116 ¶p047; 0117 ¶p010; 0120 ¶p238; 0126 ¶p211 [4] 0005 ¶p324; 0013 ¶p120; 0017 ¶p040; 0032 ¶p101; 0035 ¶p076; 0052 ¶p131; 0055 ¶p237; 0056 ¶p030; 0062 ¶p037; 0066 ¶p051; 0071 ¶p026; 0076 ¶p063; 0090 ¶p026; 0093 ¶p029; 0095 ¶p044; 0100 ¶p062; 0105 ¶p029; 0106 ¶p017; 0110 ¶p003; 0115 ¶p035; 0118 ¶p057; 0120 ¶p239; 0130 ¶p011 [5] 0075 ¶p044; 0115 ¶p034 [6] 0005 ¶p324; 0013 ¶p125; 0028 ¶p019; 0075 ¶p045; 0090 ¶p028; 0095 ¶p044; 0105 ¶p030 [7] 0003 ¶p055; 0005 ¶p324; 0013 ¶p124; 0025 ¶p237; 0033 ¶p030; 0052 ¶p132; 0055 ¶p042, ¶p043; 0076 ¶p016; 0086 ¶p009; 0090 ¶p024; 0093 ¶p029; 0100 ¶p062; 0102 ¶p010; 0105 ¶p030; 0113 ¶p038; 0116 ¶p047; 0117 ¶p010; 0118 ¶p057; 0120 ¶p240 [8] 0055 ¶p239; 0075 ¶p045; 0093 ¶p029 [9] 0033 ¶p031; 0055 ¶p238; 0071 ¶p029; 0115 ¶p036 [10] 0033 ¶p030; 0049 ¶p077; 0075 ¶p042; 0105 ¶p030 [11] 0111 ¶p036 [12] 0130 ¶p011 [13] 0062 ¶p037; 0066 ¶p045 [14] 0111 ¶p033 [15] 0095 ¶p043 [16] 0131 ¶p051, ¶p054 [17] 0066 ¶p049 [18] 0127 ¶p040 [19] 0042 ¶p041 [20] 0126 ¶p212

Q21: Shared ecosystem roles and accountability

41 comments, 90 tags: 1 support, 89 conditional

All but one of the 90 points accept FDA's shared-ecosystem idea with changes; the other agrees as proposed. Most commenters want the manufacturer to stay accountable for the device, with clinicians, institutions, societies and others supplying signal, data or expertise rather than taking over responsibility [1]. Many want each party's duties spelled out [2], several tying them to what each party knows or controls [3], and some warn against shifting responsibility onto clinicians, such as pathologists signing out results they cannot check [4].

Commenters differ on how much health systems should do. Some want them to validate devices locally [5]; others would limit them to inventories, logs, permissions and reporting, not model

validation [6], and one wants manufacturers' model surveillance kept separate from laboratories' local quality control [7]. Two warn that many health systems, especially smaller or rural ones, lack the capacity monitoring presumes [8]. One wants it made clear that monitoring or configuring a device within its validated envelope does not make a hospital a manufacturer [9]; another, that institutions be told of foundation-model changes [10].

Several see societies and standards bodies defining measures, failure taxonomies, reviewer calibration or common formats [11], though one says societies should not be required to review manufacturer data [12]. Some propose pooled or multi-site monitoring networks, or building on society-run registries [13], and others suggest roles for health plans or coordination with CMS [14]. Others ask that clinical engineering and service staff be named among stakeholders [15], that shared roles rest on a tamper-evident record of who reviewed what [16], or that a device be suspended if meaningful monitoring cannot be done [17]. One, drawing on remediation work, warns that firms often treat outsourcing as handing off responsibility rather than work [18]; another offers a drafting test asking whether each duty names its bearer, whether any satisfaction clause acts on proof rather than on the duty itself, and whether its trigger is neutral to the outcome's direction [19].

Sources. [1] 0003 ¶p057; 0005 ¶p325; 0009 ¶p043; 0013 ¶p126; 0017 ¶p039; 0025 ¶p248; 0032 ¶p101; 0033 ¶p035; 0035 ¶p078; 0042 ¶p054; 0045 ¶p084; 0049 ¶p082; 0050 ¶p017; 0052 ¶p133; 0053 ¶p011; 0055 ¶p240; 0056 ¶p025; 0075 ¶p026; 0076 ¶p037; 0083 ¶p005; 0086 ¶p009; 0087 ¶p041; 0092 ¶p010; 0093 ¶p030; 0099 ¶p015; 0100 ¶p063; 0105 ¶p032; 0113 ¶p039; 0116 ¶p039; 0118 ¶p060; 0119 ¶p037; 0120 ¶p242; 0127 ¶p040; 0130 ¶p012 [2] 0003 ¶p057; 0017 ¶p051; 0025 ¶p249; 0045 ¶p084; 0047 ¶p012; 0049 ¶p084; 0055 ¶p242; 0075 ¶p027; 0087 ¶p041; 0092 ¶p010; 0105 ¶p032; 0113 ¶p039; 0120 ¶p121, ¶p245; 0123 ¶p020; 0130 ¶p012 [3] 0017 ¶p007; 0025 ¶p263; 0049 ¶p082; 0092 ¶p009; 0130 ¶p012 [4] 0017 ¶p051; 0083 ¶p005; 0113 ¶p039; 0120 ¶p245 [5] 0017 ¶p051; 0052 ¶p090; 0087 ¶p019; 0120 ¶p108 [6] 0050 ¶p017; 0119 ¶p039; 0130 ¶p012 [7] 0083 ¶p007 [8] 0050 ¶p013; 0094 ¶p029 [9] 0100 ¶p063 [10] 0050 ¶p011 [11] 0033 ¶p033; 0042 ¶p054; 0055 ¶p244; 0056 ¶p025; 0093 ¶p030; 0119 ¶p040; 0120 ¶p243 [12] 0118 ¶p061 [13] 0021 ¶p018; 0055 ¶p246; 0116 ¶p015 [14] 0009 ¶p042; 0094 ¶p030; 0116 ¶p053 [15] 0032 ¶p103 [16] 0046 ¶p005 [17] 0118 ¶p062 [18] 0099 ¶p016 [19] 0119 ¶p061, ¶p062, ¶p063, ¶p064

Q22: Scaling evidence for modifications; what a PCCP can cover

40 comments, 68 tags: 2 support, 66 conditional

All but two of the 68 points accept the idea with changes; two agree as proposed [1]. Most commenters want re-testing scaled to a change's likely effect on clinical behavior and safety [2], several noting that a technically small change can matter more than a large one [3]. One asks instead whether the locked benchmark suite, which should never change in lower tiers, still validly tests the changed device [4].

Several sketch tiers: changes that cannot affect clinical output stay in the quality system; bounded ones, such as prompt, retrieval or guardrail changes, are re-tested, often under a PCCP; larger ones, such as to intended use, autonomy or tools, need broad revalidation or FDA review [5]. They differ on model updates: one would keep same-family, same-provider versions in the quality system after full re-benchmarking [6], another sees foundation-model versions that pass a prespecified regression battery as a natural PCCP category [7], two presume a new model version needs full or broad re-benchmarking [8], and one would bar open-ended foundation-model changes from PCCPs [9].

Several want each change tied to the benchmark elements or competencies it could affect, some setting that mapping in advance [10], while several would re-test safety behavior after any change that could touch it, however minor [11]. Several want prompt, retrieval, guardrail and orchestration changes under change control, assessed across the deployed configuration [12]. Some want the baseline's datasets, prompts or adjudicators pinned and versioned [13]; one, that the evaluation's own variability be measured, such as by re-running the unchanged device against itself [14]. Two want cumulative changes assessed [15]; single commenters add compression as a named change category [16] and sensor-to-

output verification after maintenance or change [17].

Sources. [1] 0124 ¶p004; 0127 ¶p024 [2] 0005 ¶p327; 0025 ¶p350; 0035 ¶p082; 0039 ¶p005; 0045 ¶p086; 0049 ¶p086; 0052 ¶p135; 0055 ¶p248; 0058 ¶p043; 0067 ¶p032; 0071 ¶p031; 0074 ¶p030; 0075 ¶p032; 0087 ¶p031; 0090 ¶p035; 0091 ¶p056; 0092 ¶p017; 0093 ¶p031; 0105 ¶p034; 0113 ¶p040; 0114 ¶p033; 0118 ¶p066; 0120 ¶p247 [3] 0025 ¶p353; 0035 ¶p082; 0045 ¶p085; 0052 ¶p135; 0071 ¶p031 [4] 0013 ¶p130, ¶p135 [5] 0003 ¶p060, ¶p061, ¶p062; 0005 ¶p327; 0013 ¶p131, ¶p132, ¶p133; 0045 ¶p086; 0049 ¶p087; 0055 ¶p249, ¶p250, ¶p252; 0093 ¶p031; 0113 ¶p040; 0120 ¶p248, ¶p249, ¶p250 [6] 0013 ¶p132 [7] 0100 ¶p068 [8] 0090 ¶p035; 0091 ¶p052 [9] 0118 ¶p066 [10] 0035 ¶p082; 0067 ¶p032; 0090 ¶p035; 0091 ¶p056; 0114 ¶p032 [11] 0005 ¶p299; 0058 ¶p043; 0071 ¶p032; 0091 ¶p057; 0118 ¶p066 [12] 0075 ¶p032; 0090 ¶p034; 0099 ¶p021; 0105 ¶p034; 0127 ¶p036 [13] 0085 ¶p035; 0090 ¶p036; 0091 ¶p059 [14] 0091 ¶p021, ¶p033 [15] 0052 ¶p135; 0075 ¶p032 [16] 0027 ¶p036 [17] 0032 ¶p074

Q23: Adapting PCCPs to changes that can't be fully prespecified

19 comments, 27 tags: 26 conditional, 1 neutral

Twenty-six of the 27 points accept FDA's approach with changes; the other describes the commenter's own governance framework [1]. Most would have a PCCP prespecify the bounds, tests or properties any change must meet, rather than list the changes [2]. Two frame this as invariants [3], two as performance bounds around the baseline [4], and four as a prespecified verification protocol or gate [5].

Two warn the plan must not become open-ended authorization [6], one that the benchmark suite must be escrowed [7], and one that a performance envelope cannot replace defining the scope of change, since averages can hide subgroup decline [8]. One suggests devices could gain independence in stages and lose it on postmarket signals [9]. For adaptive devices, one wants the permitted scope of adaptation defined [10], another no silent learning in production [11]. Another would resume reliance on a withdrawn model version only after recording what was re-established [12].

Sources. [1] 0016 ¶p003 [2] 0005 ¶p328; 0013 ¶p022; 0016 ¶p002; 0025 ¶p358; 0035 ¶p083; 0045 ¶p088; 0049 ¶p088; 0051 ¶p024; 0052 ¶p137; 0058 ¶p042; 0087 ¶p034; 0093 ¶p032; 0105 ¶p035; 0118 ¶p068; 0120 ¶p253 [3] 0013 ¶p138; 0025 ¶p358 [4] 0016 ¶p004; 0113 ¶p041 [5] 0051 ¶p024; 0058 ¶p042; 0093 ¶p032; 0118 ¶p068 [6] 0105 ¶p035; 0120 ¶p255 [7] 0013 ¶p148 [8] 0116 ¶p045 [9] 0052 ¶p138 [10] 0075 ¶p035 [11] 0120 ¶p112 [12] 0131 ¶p062

Q24: Changes initiated by third-party foundation model developers

44 comments, 84 tags: 1 support, 83 conditional

All but one of the 84 points accept FDA's framing with changes; the other agrees that the paper's evaluation of the assembled device is the baseline for re-checking such changes [1]. Several stress that the manufacturer stays responsible for its device whatever the model developer does [2].

Most want contractual and technical controls combined [3]. Proposed contract terms include advance notice, version identification, deprecation periods, access to earlier versions and rollback support [4]; suggested notice runs from 14 days with a test sandbox to 90 days [5]. Many want model versions pinned, often where feasible [6], several as a design expectation or condition of authorization [7], and many want no new version to reach patients until it passes re-benchmarking or regression tests [8].

Some call notice necessary but not enough, since, for example, it states intent rather than effect or cannot be verified in real time [9], and several want direct detection, such as fixed probe sets run against the live model [10]. One notes that pinning may be unavailable, ends in forced migration and misses changes beneath a stable label [11]; another would treat forced migration as a new-submission change [12]. When a provider will not allow pinning, notice or enough control, several call that a risk or deficiency in itself, to be disclosed, justified, or offset by narrower use or more premarket evidence [13].

On PCCPs, some would use them for pinned, re-benchmarked upgrades or anticipated changes [14], while one calls a PCCP the wrong instrument for changes a sponsor cannot predetermine [15] and

another says it cannot replace supplier controls or evaluation of the assembled device [16]. Other suggestions include hash-verified weights on the device or locally hosted models [17], keeping safety-critical logic outside the model [18], a per-inference record of which model actually answered, after one commenter found its own pipeline calling a vendor other than the one documented [19], telling deployers or users of changes [20], and support commitments so retired models do not strand working equipment [21].

Sources. [1] 0119 ¶p054 [2] 0017 ¶p041; 0021 ¶p019; 0025 ¶p265; 0032 ¶p057; 0045 ¶p090; 0113 ¶p042 [3] 0003 ¶p065; 0005 ¶p329; 0013 ¶p153; 0017 ¶p041; 0018 ¶p038; 0021 ¶p019; 0025 ¶p267; 0030 ¶p013; 0035 ¶p083; 0045 ¶p090; 0049 ¶p090; 0055 ¶p264, ¶p272; 0062 ¶p050; 0076 ¶p044; 0090 ¶p038; 0093 ¶p033; 0100 ¶p066; 0105 ¶p036; 0113 ¶p042; 0116 ¶p046; 0118 ¶p070; 0120 ¶p258; 0123 ¶p062; 0126 ¶p213; 0130 ¶p013 [4] 0013 ¶p152; 0017 ¶p041; 0021 ¶p019; 0035 ¶p083; 0055 ¶p264, ¶p265, ¶p268, ¶p269, ¶p271; 0100 ¶p066; 0113 ¶p042; 0120 ¶p259; 0123 ¶p062 [5] 0003 ¶p064; 0102 ¶p010; 0117 ¶p009 [6] 0003 ¶p065; 0006 ¶p027; 0013 ¶p151; 0018 ¶p037; 0035 ¶p083; 0048 ¶p009; 0049 ¶p091; 0053 ¶p010; 0055 ¶p273; 0062 ¶p042; 0076 ¶p044; 0090 ¶p040; 0096 ¶p004; 0100 ¶p066; 0105 ¶p036; 0116 ¶p046; 0120 ¶p259; 0123 ¶p062; 0126 ¶p213; 0130 ¶p013 [7] 0006 ¶p027; 0013 ¶p151; 0062 ¶p042; 0076 ¶p044; 0126 ¶p213 [8] 0003 ¶p065; 0005 ¶p329; 0006 ¶p027; 0053 ¶p010; 0054 ¶p105; 0055 ¶p275; 0076 ¶p044; 0093 ¶p033; 0100 ¶p067; 0102 ¶p010; 0105 ¶p036; 0117 ¶p009; 0118 ¶p070; 0120 ¶p262 [9] 0013 ¶p153; 0062 ¶p050; 0090 ¶p039; 0131 ¶p061 [10] 0005 ¶p329; 0013 ¶p153; 0048 ¶p009; 0049 ¶p090; 0090 ¶p041; 0093 ¶p033; 0096 ¶p011; 0124 ¶p012; 0131 ¶p061 [11] 0090 ¶p040 [12] 0013 ¶p154 [13] 0003 ¶p065; 0013 ¶p151; 0021 ¶p019; 0048 ¶p009; 0055 ¶p280; 0075 ¶p038; 0076 ¶p044; 0120 ¶p261 [14] 0006 ¶p027; 0062 ¶p042; 0120 ¶p260; 0126 ¶p214 [15] 0076 ¶p044 [16] 0119 ¶p056 [17] 0082 ¶p034; 0096 ¶p007 [18] 0100 ¶p065 [19] 0062 ¶p015, ¶p040 [20] 0110 ¶p005; 0118 ¶p070; 0130 ¶p013 [21] 0032 ¶p036

MASTER_FILES: Foundation model master files

38 comments, 83 tags: 7 support, 71 conditional, 3 oppose, 2 neutral

Q25: Voluntary Foundation Model Master Files

38 comments, 83 tags: 7 support, 71 conditional, 3 oppose, 2 neutral

Of the 83 points, 71 accept master files with changes, seven support them as proposed, three oppose a voluntary program and two take no position. Many doubt that developers will volunteer safety-relevant information or keep files current [1]. Responses differ. Some would make disclosure mandatory, at least for higher-consequence devices, or a condition of the submission [2]; others would keep it voluntary but create pressure through device makers, with heavier evidence burdens when no adequate file exists [3] or credit, such as more predictable review or a lighter benchmarking burden, when one is referenced [4]. One instead wants no presumption against devices without a file, provided the sponsor characterizes the model itself [5].

Many stress, in line with the paper, that a file cannot authorize the model, replace validation of the final device or shift the manufacturer's responsibility [6]; one warns that reviewers may still over-trust a file describing the model before fine-tuning, compression and prompting [7].

Many list contents, such as version history, data provenance, known limitations and failure modes, subgroup results, guardrails and update-notification commitments [8]. Additions include deprecation and rollback terms [9], refusal behavior [10], safety under downstream changes such as quantization [11], where the model actually runs [12], and the developer's own detection methods, since a file can disclose only what the developer knows [13]. Several stress currency [14]; one, having measured a hosted assistant's model swapped 153 times in fourteen hours, would limit files to stable facts and record the answering model continuously [15].

On confidentiality, two want a public tier or public model cards [16], one tiered disclosure to sponsors [17], and another hash commitments to results that developers keep [18]. Single commenters suggest letting cloud platforms hold files [19], a fillable template [20], expecting more of closed models than open-weight ones [21], and reaching model information through manufacturers' purchasing requirements [22], noting that rules on AI in coverage decisions place duties on the user, not the model developer [23].

Sources. [1] 0003 ¶p068; 0013 ¶p158; 0017 ¶p043; 0045 ¶p092; 0076 ¶p039; 0093 ¶p034; 0100 ¶p071; 0102 ¶p010; 0106 ¶p018; 0113 ¶p044; 0115 ¶p043; 0117 ¶p009; 0118 ¶p072; 0119 ¶p055; 0131 ¶p064 [2] 0076 ¶p040; 0106 ¶p018; 0113 ¶p044 [3] 0003 ¶p069; 0013 ¶p160 [4] 0006 ¶p027; 0058 ¶p047; 0118 ¶p072 [5] 0100 ¶p071 [6] 0005 ¶p330; 0013 ¶p164; 0025 ¶p368; 0032 ¶p102; 0046 ¶p020; 0049 ¶p095; 0054 ¶p096; 0055 ¶p132; 0087 ¶p038; 0099 ¶p017; 0105 ¶p038; 0114 ¶p026; 0115 ¶p043; 0119 ¶p055; 0120 ¶p268; 0126 ¶p217 [7] 0027 ¶p067, ¶p068 [8] 0003 ¶p070; 0005 ¶p330; 0013 ¶p162; 0017 ¶p043; 0035 ¶p085; 0049 ¶p094; 0054 ¶p096; 0055 ¶p120, ¶p123, ¶p128; 0087 ¶p037; 0093 ¶p034; 0105 ¶p037; 0113 ¶p044; 0115 ¶p042; 0120 ¶p265 [9] 0032 ¶p102; 0035 ¶p085; 0115 ¶p042 [10] 0058 ¶p045 [11] 0027 ¶p071 [12] 0079 ¶p035 [13] 0008 ¶p010, ¶p011 [14] 0005 ¶p330; 0013 ¶p163; 0017 ¶p043; 0049 ¶p093; 0105 ¶p037; 0120 ¶p267 [15] 0079 ¶p005, ¶p032 [16] 0050 ¶p016; 0076 ¶p041 [17] 0035 ¶p085 [18] 0131 ¶p065 [19] 0100 ¶p072 [20] 0093 ¶p034 [21] 0131 ¶p068 [22] 0119 ¶p052 [23] 0119 ¶p047

AGENTIC: Agentic AI systems

52 comments, 148 tags: 4 support, 144 conditional

Q26: Agentic AI systems

52 comments, 148 tags: 4 support, 144 conditional

All but four of the 148 points accept FDA's approach with changes. The four agree as proposed with the agentic competencies in benchmarking element A.1 [1], with full device oversight for agentic sequences that control another device [2], and with the paper's view that agentic care coordination, documentation and outreach may fall outside device focus [3].

Many say that because agents act, evaluation must cover the whole chain of actions and its effects, not just the final output [4], several warning that small errors compound across steps [5]; one notes that 95 percent accuracy per step gives about 60 percent over ten steps [6].

Most want a human checkpoint before irreversible or high-consequence actions [7]. Several add that a checkpoint must be real: enforced outside the model, not bypassable by wording, inattention or the agent's own reasoning, not satisfiable by another model, giving the reviewer enough information, and more than a notification [8]; one notes that small and rural sites may lack staff for such review [9], and another wants a tamper-evident record that a present human authorized each consequential act [10]. Several, though, would allow autonomous execution that has been specifically evaluated and authorized [11].

Many want a declared set of permitted tools, data and actions, with least privilege [12], several stressing that capability does not confer authority to act [13]. Several would scale criteria or checkpoints to autonomy, consequence and reversibility [14], proposing tools such as an action inventory classed by reversibility [15], a cap on consecutive autonomous actions [16], a five-level agency ladder [17] or permission tiers [18].

Many want logs that let investigators reconstruct what an agent did [19], and some specify records the agent cannot alter, stating who authorized each act and showing skipped checkpoints or unknown actors as such [20]. Since users see only the agent's own account of its steps, one wants that account checked against an independent execution record [21]. Several want a stop control, rollback or safe fallback [22].

Several want testing for prompt injection, including through retrieved content and tool outputs [23], and of tool failures and partial execution [24]. Single commenters would also test for authority gained mid-session [25], memory an agent writes about its user [26], and approval steps with intended users [27]. One calls element A.1 "woefully inadequate" [28]; another, while supporting its competencies, lists properties it misses, such as whether step reports are true [29].

On scope, two want agentic architecture alone not to make a non-device function a device [30], while one warns that administrative agents outside FDA's focus can still shape care and suggests cross-agency coordination may be needed [31]. Single commenters would judge an agent against the risk of any device it controls [32], keep agents out of the reduced-premarket-evidence pathway [33], monitor them more often [34], or set up a separate working group on surgical robots [35]. One wants clearly defined responsibility for the end-to-end system when components come from different providers [36]; another, the agent, models, tools and datasets registered as a pinned configuration whose changes

take effect only once prespecified evidence exists [37]. A third warns that nonverbal users cannot contest what an agent says or stores about them [38].

Sources. [1] 0061 ¶p042 [2] 0093 ¶p035 [3] 0094 ¶p069; 0100 ¶p073 [4] 0005 ¶p331; 0013 ¶p166; 0018 ¶p030; 0025 ¶p275; 0035 ¶p087; 0048 ¶p010; 0049 ¶p097; 0052 ¶p145; 0056 ¶p027; 0071 ¶p028; 0075 ¶p011; 0087 ¶p047; 0105 ¶p039; 0113 ¶p045; 0115 ¶p044; 0120 ¶p272 [5] 0013 ¶p166; 0035 ¶p089; 0049 ¶p099; 0052 ¶p145; 0055 ¶p136; 0056 ¶p027; 0067 ¶p033; 0095 ¶p046; 0118 ¶p075 [6] 0013 ¶p166 [7] 0003 ¶p073; 0013 ¶p168; 0017 ¶p045; 0018 ¶p031; 0035 ¶p089; 0046 ¶p020; 0048 ¶p010; 0049 ¶p099; 0052 ¶p148; 0053 ¶p012; 0054 ¶p099; 0055 ¶p147; 0056 ¶p028; 0067 ¶p033; 0076 ¶p063; 0087 ¶p046; 0089 ¶p008; 0092 ¶p019; 0094 ¶p068; 0097 ¶p006; 0098 ¶p045; 0100 ¶p074; 0101 ¶p006; 0102 ¶p010; 0105 ¶p040; 0113 ¶p046; 0115 ¶p046; 0117 ¶p010; 0118 ¶p074; 0120 ¶p273; 0127 ¶p042 [8] 0048 ¶p010; 0056 ¶p028; 0067 ¶p033; 0075 ¶p012; 0094 ¶p068; 0098 ¶p045; 0100 ¶p074; 0102 ¶p010; 0111 ¶p021; 0117 ¶p010; 0127 ¶p042 [9] 0094 ¶p068 [10] 0089 ¶p002 [11] 0035 ¶p089; 0054 ¶p099; 0087 ¶p046; 0105 ¶p040; 0120 ¶p273 [12] 0013 ¶p169; 0017 ¶p045; 0035 ¶p088; 0049 ¶p099; 0055 ¶p144, ¶p145; 0056 ¶p028; 0075 ¶p012; 0076 ¶p063; 0093 ¶p035; 0101 ¶p006; 0105 ¶p040; 0115 ¶p046; 0122 ¶p026; 0130 ¶p014 [13] 0029 ¶p018; 0045 ¶p124; 0049 ¶p103; 0054 ¶p097; 0087 ¶p045; 0092 ¶p019; 0122 ¶p025 [14] 0003 ¶p074; 0027 ¶p079; 0045 ¶p094; 0051 ¶p025; 0052 ¶p148; 0054 ¶p009; 0089 ¶p017; 0093 ¶p035; 0097 ¶p006; 0118 ¶p074 [15] 0013 ¶p168; 0067 ¶p034 [16] 0003 ¶p074 [17] 0045 ¶p094 [18] 0093 ¶p035; 0122 ¶p026 [19] 0013 ¶p171; 0017 ¶p045; 0018 ¶p031; 0024 ¶p008; 0049 ¶p099; 0054 ¶p099; 0056 ¶p029; 0067 ¶p034; 0075 ¶p048; 0076 ¶p063; 0093 ¶p035; 0113 ¶p046; 0120 ¶p272; 0130 ¶p014 [20] 0042 ¶p047, ¶p048; 0075 ¶p049; 0127 ¶p042; 0131 ¶p072, ¶p073 [21] 0061 ¶p015, ¶p049 [22] 0003 ¶p075; 0017 ¶p045; 0035 ¶p089; 0052 ¶p148; 0055 ¶p150; 0100 ¶p076; 0109 ¶p013; 0113 ¶p046; 0115 ¶p046; 0118 ¶p076; 0130 ¶p014 [23] 0005 ¶p331; 0035 ¶p088; 0048 ¶p010; 0055 ¶p140; 0056 ¶p029; 0075 ¶p015; 0100 ¶p075; 0115 ¶p046; 0120 ¶p272 [24] 0035 ¶p089; 0048 ¶p010; 0056 ¶p029; 0075 ¶p016; 0100 ¶p076; 0120 ¶p272 [25] 0071 ¶p034 [26] 0061 ¶p051 [27] 0075 ¶p013 [28] 0043 ¶p018 [29] 0061 ¶p044 [30] 0094 ¶p070; 0120 ¶p275 [31] 0025 ¶p287, ¶p296 [32] 0051 ¶p025 [33] 0013 ¶p172 [34] 0056 ¶p030 [35] 0003 ¶p076 [36] 0052 ¶p144 [37] 0122 ¶p009, ¶p015 [38] 0061 ¶p063

MISC: Outside the paper's topics

28 comments, 65 tags: 65 neutral

X-PRIV: Privacy and data protection

7 comments, 14 tags: 14 neutral

All 14 points take no position on FDA's proposals. Three worry that patient data used by AI may not be kept private or safe from breach, theft or sale [1]; another cautions organizations against giving AI too much sensitive data [2]. Three object to personal data, including emergency-room voice prints, being used or held without informed consent [3]; one adds that voice prints cannot be de-identified [4]. One objects to physicians entering patients' messages into a clinical AI tool [5]. Another wants FDA to ask sponsors to account for each class of data GenAI creates, with minimization, deletion and no silent reuse for training [6].

Sources. [1] 0007 ¶p001; 0019 ¶p020; 0023 ¶p005 [2] 0110 ¶p004 [3] 0019 ¶p003; 0041 ¶p001; 0044 ¶p001, ¶p005 [4] 0044 ¶p004 [5] 0019 ¶p014 [6] 0035 ¶p070, ¶p071

X-LIAB: Liability

8 comments, 13 tags: 13 neutral

All 13 points take no position on FDA's proposals. Most want those who make, own, sell or deploy AI devices to answer legally for the harm they cause [1], three adding that clinicians or hospitals should not carry liability alone [2]. Two reject disclaimers as a way to shift blame [3], and one asks whether manufacturers or the government would answer [4]. Another argues that product liability law already holds makers strictly liable, a higher standard than doctors face [5]. One says insurers need stable review metrics and logs to price AI risk, with contracts allocating liability [6].

Sources. [1] 0011 ¶p003; 0022 ¶p003, ¶p008; 0097 ¶p008; 0113 ¶p042; 0118 ¶p063 [2] 0097 ¶p008; 0113 ¶p001; 0118 ¶p063 [3] 0019 ¶p016; 0022 ¶p006 [4] 0011 ¶p002 [5] 0031 ¶p004, ¶p009, ¶p010 [6] 0092 ¶p080, ¶p091, ¶p092

X-PAY: Reimbursement, coverage and cost

2 comments, 3 tags: 3 neutral

All three points take no position on FDA's proposals. One commenter calls the missing discussion of reimbursement and coverage the paper's most significant omission, warns payers may treat cleared devices as experimental for years, and proposes a formal FDA–CMS liaison so one body of clinical evidence serves both clearance and coverage [1]. Another, on rural stroke care, says insurers will need to accept transfers based on AI interpretation [2].

Sources. [1] 0003 ¶p012, ¶p013, ¶p079, ¶p080 [2] 0036 ¶p001

X-LABEL: Labeling and transparency to users

10 comments, 15 tags: 15 neutral

All 15 points take no position on FDA's proposals. Four want patients or other users told when AI is involved in their care or in what they see [1]; two add that disclosure must be understandable [2] or is not consent by itself [3]. Four want users given information such as a device's limitations, failure modes, subgroup performance and test results [4]. Other proposals include disclosing data collection and reuse [5], the foundation model or other providers relied on [6], and which functions need networks or

cloud services, with fallbacks [7]; a competency "nutrition label" [8]; declared write authority on model cards [9]; and a disclaimer that AI is not fail-safe [10].

Sources. [1] 0003 ¶p081; 0047 ¶p007; 0086 ¶p007; 0097 ¶p011 [2] 0047 ¶p013 [3] 0086 ¶p007 [4] 0003 ¶p081; 0017 ¶p049; 0114 ¶p019; 0118 ¶p034, ¶p073 [5] 0017 ¶p049 [6] 0003 ¶p081; 0017 ¶p049 [7] 0032 ¶p071, ¶p072 [8] 0114 ¶p018 [9] 0127 ¶p043 [10] 0019 ¶p006

X-WORK: Workforce and staff training

3 comments, 5 tags: 5 neutral

All five points take no position on FDA's proposals. Two want clinicians free to question, override or stop AI recommendations without retaliation or institutional or financial pressure [1], and both reject AI as a substitute for safe staffing, one warning that savings from fewer physicians may push autonomous AI beyond what the evidence supports [2]. A third wants defined competency standards for research staff who act on AI outputs in regulated trial work [3].

Sources. [1] 0086 ¶p006; 0097 ¶p008 [2] 0086 ¶p007; 0097 ¶p009, ¶p010 [3] 0112 ¶p013, ¶p014

X-ADOPT: Clinical adoption

2 comments, 2 tags: 2 neutral

Both points take no position on FDA's proposals. One commenter notes that internet connectivity is an issue for some families [1]. Another, writing from a rural hospital, says AI interpretation of head CT scans could improve outcomes for stroke patients at rural hospitals when no radiologist is available, but only if physicians at neighboring hospitals accept AI interpretations and the resulting transfers [2].

Sources. [1] 0014 ¶p001 [2] 0036 ¶p001

X-INTEROP: Interoperability

3 comments, 4 tags: 4 neutral

All four points take no position on FDA's proposals. All three commenters want information exchanged between connected systems to keep its provenance [1], one treating interoperability as an evidence problem for reconstructing events [2]. One wants each safety-relevant exchange to carry identity, purpose, meaning and onward-use limits, since encryption alone does not show it was authorized or valid, with stronger safeguards for control commands [3]. Another wants FDA to encourage standards-based interoperability [4].

Sources. [1] 0035 ¶p079; 0049 ¶p104; 0087 ¶p043 [2] 0049 ¶p104 [3] 0035 ¶p079, ¶p104 [4] 0087 ¶p043

X-OTHER: Other topics outside the paper

4 comments, 9 tags: 9 neutral

All nine points take no position on FDA's proposals. One commenter wants technicians and independent servicers given the manuals, diagnostics and access to maintain devices, without cybersecurity blocking them [1], and pre-purchase disclosure of functions needing continuing payment or open to remote disabling [2]. Another wants authorization to include a discontinuation plan, after two mental health apps vanished without notice [3]. A third wants patients able to request human review and correct wrong information [4]; a fourth raises ethical questions about personhood and how device parameters are tuned [5].

Sources. [1] 0032 ¶p016, ¶p017, ¶p061, ¶p063 [2] 0032 ¶p083 [3] 0021 ¶p017 [4] 0047 ¶p013 [5] 0004 ¶p001, ¶p002

MISC: Outside the paper's topics

Not raised: X-SEC, X-IP